Experiment 3: Visual-Inertial Odometry With Dynamic-Environment Filtering

Raw write-up for Chapter 4, Section 4.3

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Purpose of this document. This is the raw, complete write-up of Experiment 3, written before the thesis chapter; Section 4.3 of the final report is extracted from it. It holds both result slots, every table and figure, and the analysis. Every number traces to a file under `output/` named in the "Reporting rules" paragraph of its testbed. Open concerns are set in red where they arise.

4.3.1 Simulation

Status: partial. Two testbeds, analysed separately. The extreme world (scripted `small_warehouse_dynamic_*_nofloortexture` warehouses, 66 September runs and 14 October filtered runs) left the filtering hypothesis untested: the VINS-Fusion filter never activated in September, eleven of twelve ORB-SLAM3 treatment pairs were not input-matched, and the testbed itself is not physical. The realistic world (physically simulated traffic, 32 runs of 14–21 October 2026) gives the environment-dynamics factor its first valid test from one run per condition; its twenty filtered runs yield no admissible pair.

Objective. Determine whether masking dynamic-object features improves localisation, at what compute cost, and whether tracking survives it.

Why two testbeds. The two environments differ in what "dynamic" means, so their results are not pooled (table 3.1). In the extreme world the props follow scripted routes through a kinematic plugin: they move continuously through the camera's view and carry no physical response. In the realistic world the moving objects are pallet jacks on logistics jobs and people on foot, in two layouts, each with a static twin that differs in traffic only.

Table 3.1: The two simulation testbeds of Experiment 3. Both use the live detector at replay time.

	Extreme world	Realistic world
Worlds	`small_warehouse_dynamic_{00,01,02}_nofloortexture`; static controls `small_warehouse_static[_01]_nofloortexture`	`small_warehouse_dynamic_realistic_nofloortexture` (U) and `_01` (I); static twins of each
Moving objects	scripted kinematic props, continuous motion, no physical response	four pallet jacks on pick–carry–drop jobs, four walking people; 10 and 12 props move during the recording
Recordings	3 static, 4 dynamic (`*_nofloortexture_*`)	2 static, 2 dynamic (`allsensor_realistic_000–003`)
Runs	66 September repeats (both pipelines, baseline and class mask, three per condition) plus 14 October filtered runs	32: 12 baseline (VINS-Fusion; ORB-SLAM3 with and without loop closer), 20 filtered (class mask, motion gate; 8 with a prior atlas)
Replay	0.5×, 16 September 2026	1.0×, 14–21 October 2026
Standing	dynamic trajectories withheld; two comparable treatment pairs of twenty-four, in an invalid testbed	baselines valid; no admissible treatment pair

4.3.1.1 Extreme world: scripted dynamic warehouses

Objective. Measure both pipelines, unfiltered and class-masked, on warehouses whose props move continuously through the camera's view.

Process. The relogged campaign of 16 September 2026 ran both pipelines on seven featureless-floor recordings, baseline and class-mask treatments as separate processes on identical input at 0.5× replay, three repeats per condition. Fourteen further filtered runs were made in October 2026 on the same recordings at replay rate 1.0 with the live detector, adding the motion gate and an ORB-SLAM3 arm with its loop closer disabled.

Datasets. Three static and four dynamic `nofloortexture` recordings (table 3.1); routes are drawn in every trajectory figure below.

Reporting rules. Fixed on 21 October 2026, before the tables were read. Unfiltered arms: the *best of three repeats* per recording, best being the lowest aligned ATE, with median and worst ATE beside it. Filtered arms: every run outside the September repeat set, *one by one*, never as treatment pairs, because their baselines are a different session, node build and replay rate. Real-time and resource figures: mean ± SD over the three September repeats. Alignment is a rigid $SE(3)$ fit with scale fixed to the rosbag ground truth; jump-free RPE excludes pairs spanning a step above $10\,\mathrm{m\,s^{-1}}$; the rotational RPE is quoted as the median because ORB-SLAM3's single 90° re-alignment at inertial initialisation inflates any RMSE. Estimates are scored against the ground truth moved from the model origin to the IMU point both estimators report, a lever arm of 0.338 m forward and 0.192 m up taken from the robot model (`models/ackermann_robot/model.sdf`), and every figure draws that moved ground-truth path. The campaign was re-analysed from the run tree on 21 October; two September runs on `dynamic_00_000` repeat 1 had been overwritten by 15 October re-runs, and under the re-analysis the campaign holds two comparable treatment pairs, one per estimator, both in this invalid testbed. Sources: `output/compare/relogged_20260916/`, `output/compare/simulation/exp3-rules/`, `output/compare/simulation/exp3-timelines/`.

Validity. All 66 September runs are operational, but the campaign holds no valid test of the filter (table 3.2). The VINS-Fusion filter never activated: every nominal VINS+YOLO run reports zero matched detector frames and zero mask coverage. The ORB-SLAM3 filter ran, but eleven of twelve baseline/filter pairs differ in received-frame count by 10–30 %, outside the 5 % pairing rule. Every one of the 48 dynamic runs contains impossible-speed steps, with a median affected-step fraction of 60 % (VINS-Fusion) and 30 % (ORB-SLAM3), so a global ATE in this testbed is not ordinary drift. The anomaly catalogue of table 3.3 records what the logs show without assigning causes; figure 3.1 draws it.

Table 3.2: September campaign, progress and validity. "Operational" requires graceful shutdown, initialisation, at least ten processed frames and a trajectory; a valid treatment pair additionally requires an active filter and paired input counts within 5 %.

Run family	Attempted	Operational	Valid pairs	Note
Static baseline	18	18	–	two ORB-SLAM3 run summaries incomplete; frame logs confirm initialised operation
Dynamic baseline	24	24	–	every run has impossible-speed steps
Dynamic ORB-SLAM3+YOLO	12	12	1 of 12	eleven pairs input-mismatched
Dynamic VINS-Fusion+YOLO	12	12	0 of 12	filter inactive in every run
Total	66	66	1 (2 after re-analysis)	

Table 3.3: Anomaly catalogue of the September campaign. Observations only; no root cause is assigned.

ID	Severity	Observation and consequence
A1	Critical	VINS-Fusion+YOLO filtering inactive: 28 038 mask rows, zero matched detector messages. Cannot measure a treatment.
A2	High	ORB-SLAM3+YOLO exposure varies between repeats: positive-box fraction 1.4–99.2 %. Match count is not dose.
A3	Critical	Every dynamic trajectory has discontinuities: median affected-step fraction 59.6 % (VINS), 30.3 % (ORB). Global ATE is not drift.
A4	High	Global-to-clean ATE ratios of 100–239: locally coherent intervals coexist with catastrophic jumps.
A5	Medium	Static ORB-SLAM3 has one initialisation-adjacent jump near 2.6 s, preceding the inertial transition by 0.10–0.17 s.
A6	Medium	Two static ORB-SLAM3 run summaries omit final fields; retained on frame-log evidence.
A7	High	ORB-SLAM3 baseline/filter inputs unpaired in 11 of 12 cases (9.95–30.06 % frame difference, durations within 0.13 %).
A8	Medium	VINS-Fusion failure detection disabled, zero resets in all 33 runs: resets cannot signal health.

Results. Unfiltered, the best repeat on every dynamic recording is wrong by more than the robot travels, 11–2311 m aligned ATE, while the static recordings hold 0.10–0.56 m (table 3.4). Figures 3.2 and 3.3 show every repeat behind the table.

The static best-of-three is a best case, not a typical one. The medians are 0.12–0.30 m for VINS-Fusion and 0.20–0.74 m for ORB-SLAM3, and the worst ORB-SLAM3 repeat on `static_000` reaches 1.15 m.

ORB-SLAM3's rotational RMSE of about 17° on the static recordings is one event, not its tracking. Each selected run carries exactly one discontinuity near inertial initialisation; the median rotational RPE at 1 s is 0.05–0.13° for both estimators.

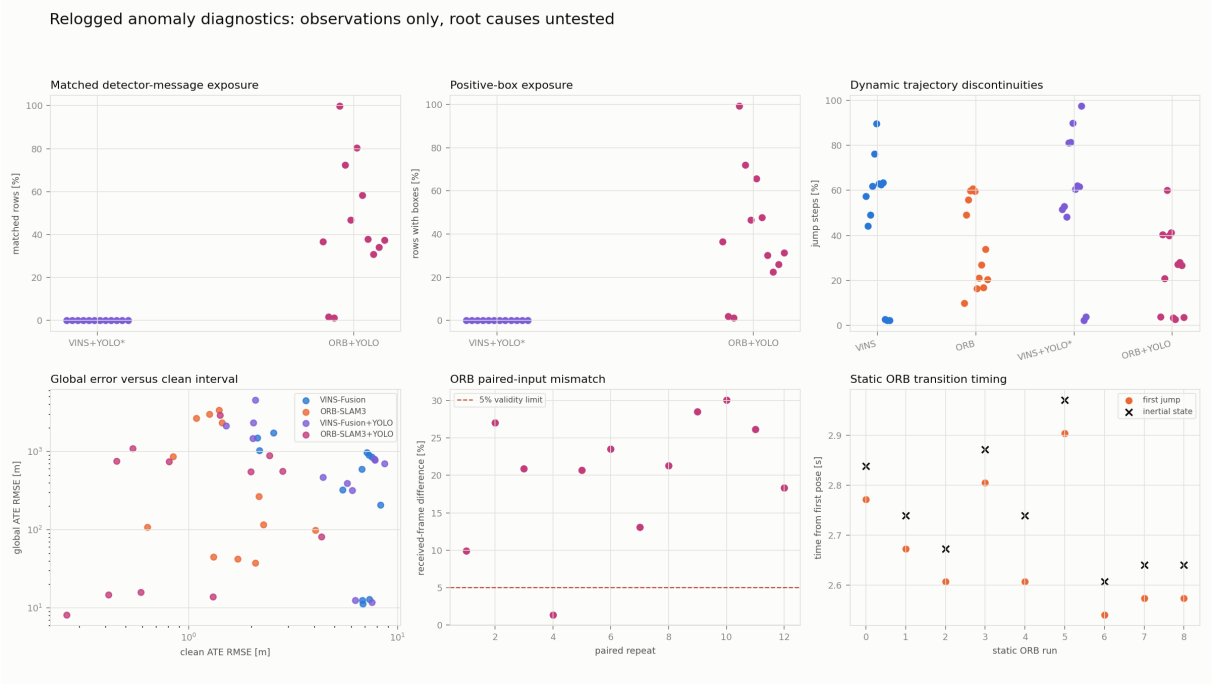


Figure 3.1: Run-level anomaly diagnostics of the September campaign: detector-message and positive-box exposure, dynamic jump fractions, global versus clean-segment ATE, ORB-SLAM3 paired-input mismatch and static ORB-SLAM3 jump timing.

The dynamic numbers are withheld as findings and listed as excluded evidence. The one run under 12 m, VINS-Fusion on `dynamic_02_000`, still carries 58 jumps.

Filtered runs, October. A filter that actually runs changes the outcome, but nothing here is a treatment effect (table 3.5, figure 3.4). The fourteen runs are unpaired and the testbed is invalid; they are recorded as the only evidence so far that a running mask alters what happens on these recordings.

Where the mask was present on most frames, ORB-SLAM3 stays within 8–20 m with 62–96 % fewer discontinuities than the best unfiltered repeat. The exception is the class mask with loop closure on `dynamic_02_000`, which diverges to 1391 m where the same mask without the closer holds 10.1 m.

VINS-Fusion separates by mask rate. The three class runs with a mask on 15–24 % of frames diverge to 1500–2900 m; the class run with a mask on 93 % of frames holds 6.4 m over 73 s on `dynamic_00_001`, the only VINS-Fusion result under 10 m on any dynamic recording of this testbed. The motion-gated run on `dynamic_00_001` overlaps the ground truth for 5 s and is not read; the four ORB-SLAM3 runs on `dynamic_00_000` cover 49 of 77 s.

Analysis: when, where, and how long the filter ran. Unfiltered, every dynamic run fails on the first aisle, before any corner (table 3.6). The error series is the position error after a rigid fit on an anchor window near the start (after the initialisation transient, until the ground truth has turned 60°); the onset is the first time it stays above three times the estimator's static median for 1 s, here 0.34 m for VINS-Fusion and 0.75 m for ORB-SLAM3. All 24 dynamic baseline runs cross, 23 on a straight, ORB-SLAM3 at 3–18 s and VINS-Fusion at 13–32 s, all but one within the first quarter of the route. That is the signature of props moving continuously through the view, not of a turn the tracker cannot follow.

A running mask delays the failure by tens of seconds but does not prevent it (table 3.7). In the September repeats the filter worked for the whole run in two, for 5–60 s in five and never in seventeen, and where it never ran the onsets fall where the baselines' do. All four October ORB-SLAM3 runs on `dynamic_00_000` survive the first two aisles and cross at 42–43 s in the south-west corner, at about

Table 3.4: Extreme world, unfiltered arms: the best of the three September repeats per recording, "best" being the lowest aligned ATE. Rotation, jump-free RPE and jump count are those of the selected run; the median and worst ATE of the three are kept beside the best so the selection is visible. Jump-free RPE excludes pairs spanning a step above $10\,\mathrm{m\,s^{-1}}$; the rotational figure is the median over pairs, because a single orientation step inflates the RMSE (section 4.3.1.2).

Recording	Estimator	Best run	Aligned ATE [m]			Rot. RMSE	$\mathrm{RPE}_{1\,\mathrm{s}}$	$\mathrm{RPE}_{1\,\mathrm{s}}$ rot.	Jumps
			best	median	worst	[°]	[m]	med. [°]	
`dynamic_00_000`	VINS-Fusion	02	206.952	323.992	1401.153	88.87	3.699	0.30	609
	ORB-SLAM3	01	45.306	857.935	3364.333	176.43	3.464	1.02	109
`dynamic_00_001`	VINS-Fusion	01	1030.396	1480.989	1724.978	60.97	2.260	1.12	1374
	ORB-SLAM3	03	2311.316	2669.741	3002.582	23.48	2.024	1.19	1098
`dynamic_01_000`	VINS-Fusion	02	851.930	904.967	965.856	54.82	2.828	0.42	1596
	ORB-SLAM3	01	98.954	107.708	115.532	169.11	2.202	2.67	239
`dynamic_02_000`	VINS-Fusion	03	11.221	12.404	12.799	4.05	0.752	0.19	58
	ORB-SLAM3	01	37.739	42.290	265.485	58.38	2.107	1.79	388
`static_000`	VINS-Fusion	01	0.254	0.295	0.325	1.34	0.032	0.07	0
	ORB-SLAM3	01	0.561	0.743	1.153	17.92	0.065	0.13	1
`static_001`	VINS-Fusion	02	0.179	0.301	0.429	1.49	0.033	0.07	0
	ORB-SLAM3	03	0.347	0.424	0.432	17.13	0.039	0.10	1
`static_002`	VINS-Fusion	03	0.111	0.122	0.132	0.76	0.034	0.05	0
	ORB-SLAM3	03	0.101	0.204	0.241	16.85	0.025	0.05	1

Table 3.5: Extreme world, filtered arms: every run outside the September repeat set, one row each (October 2026, replay rate 1.0, live detector). "Match" is `detection_match_rate`; a run below 0.98 did not filter on every frame. "Kept" is the processed fraction of received frames. Loop-closer state is read from the log. "Overlap" is the span over which estimate and ground truth coexist, against the recording's length; a run covering less than 80 % of it is a partial run and its ATE describes only that span. These runs are reported on their own, not as pairs: their baselines are the September repeats of table 3.4, a different session and node build.

Recording	Estimator	Mask	LC	Match	Kept	Overlap [s]	ATE [m]	Rot. [°]	$\mathrm{RPE}_{1\,\mathrm{s}}$ [m]	Jumps
`dynamic_00_000`	ORB-SLAM3+YOLO	class	active	0.857	64 %	**49 of 77**	8.702	87.06	0.858	4
	ORB-SLAM3+YOLO	class	disabled	0.961	78 %	**49 of 77**	8.327	83.92	0.892	7
	ORB-SLAM3+YOLO	motion	disabled	0.785	57 %	**49 of 77**	10.374	139.11	1.609	23
	ORB-SLAM3+YOLO	motion	disabled	0.868	60 %	**49 of 77**	16.618	80.88	1.830	37
	VINS-Fusion+YOLO	class	—	0.152	100 %	75 of 77	2893.915	125.50	4.599	1407
	VINS-Fusion+YOLO	motion	—	1.000	100 %	77 of 77	402.521	106.82	4.283	1142
`dynamic_00_001`	VINS-Fusion+YOLO	class	—	0.930	100 %	73 of 74	6.351	9.28	0.559	1
	VINS-Fusion+YOLO	motion	—	1.000	93 %	**5 of 74**	5.266	120.90	4.563	13
`dynamic_01_000`	ORB-SLAM3+YOLO	class	active	0.992	77 %	**65 of 85**	19.781	148.99	1.558	92
	ORB-SLAM3+YOLO	class	disabled	0.998	93 %	**65 of 85**	9.402	65.44	0.817	9
	VINS-Fusion+YOLO	class	—	0.244	100 %	75 of 85	1501.826	18.13	0.863	907
	VINS-Fusion+YOLO	motion	—	0.998	100 %	**65 of 85**	283.596	55.37	3.000	893
`dynamic_02_000`	ORB-SLAM3+YOLO	class	active	0.939	75 %	77 of 89	1391.039	96.10	1.615	441
	ORB-SLAM3+YOLO	class	disabled	0.927	79 %	77 of 89	10.142	62.43	1.316	27
	VINS-Fusion+YOLO	class	—	0.241	100 %	89 of 89	2260.213	172.26	2.694	1216
	VINS-Fusion+YOLO	motion	—	0.999	100 %	77 of 89	333.570	130.90	3.625	1087

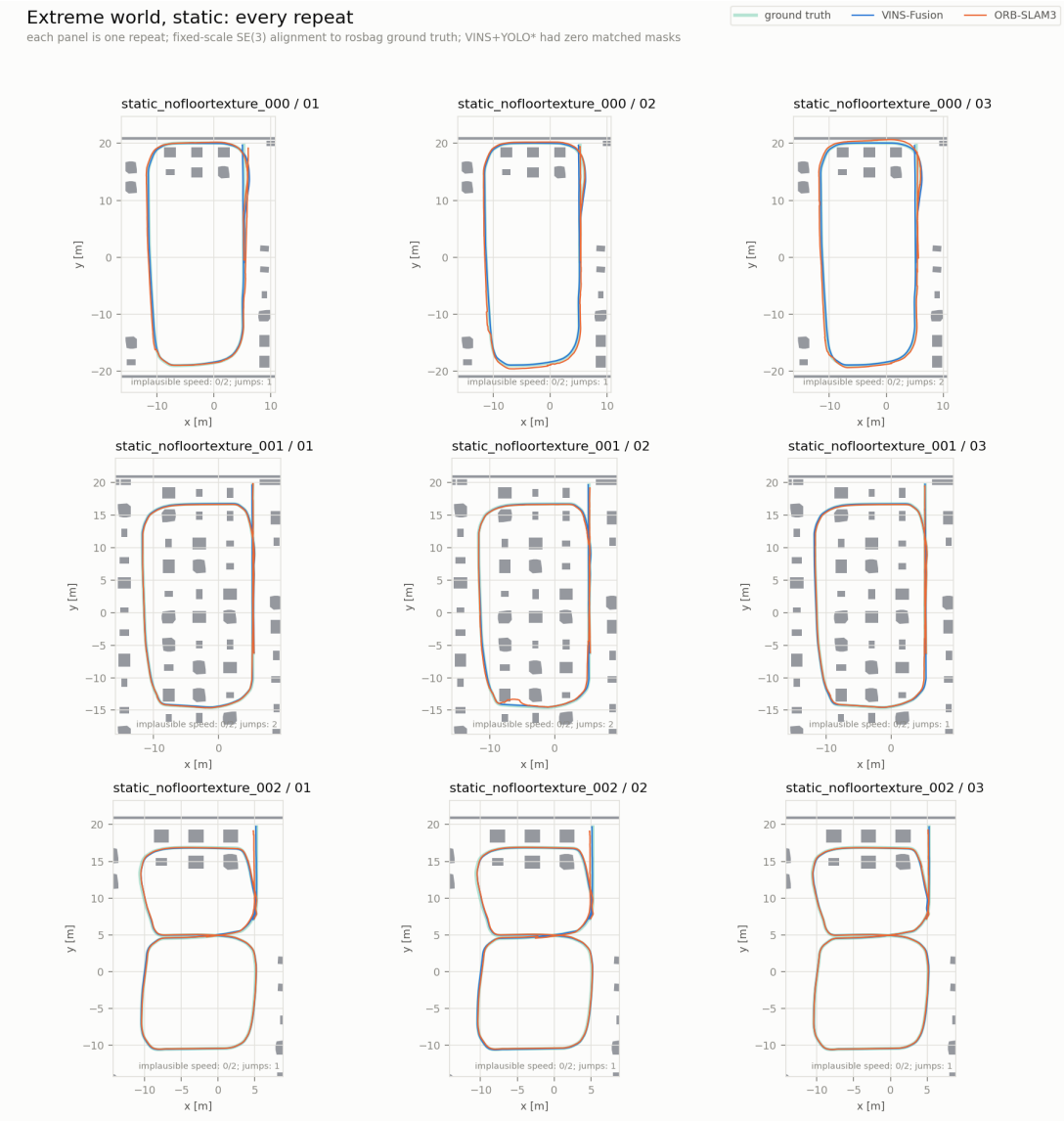


Figure 3.2: Extreme world, static recordings: every repeat of both front ends, top-down after rigid $SE(3)$ alignment to the rosbag ground truth (pale green) on the warehouse footprint. Each panel's footer counts runs with an implausible mean speed and the discontinuities found.

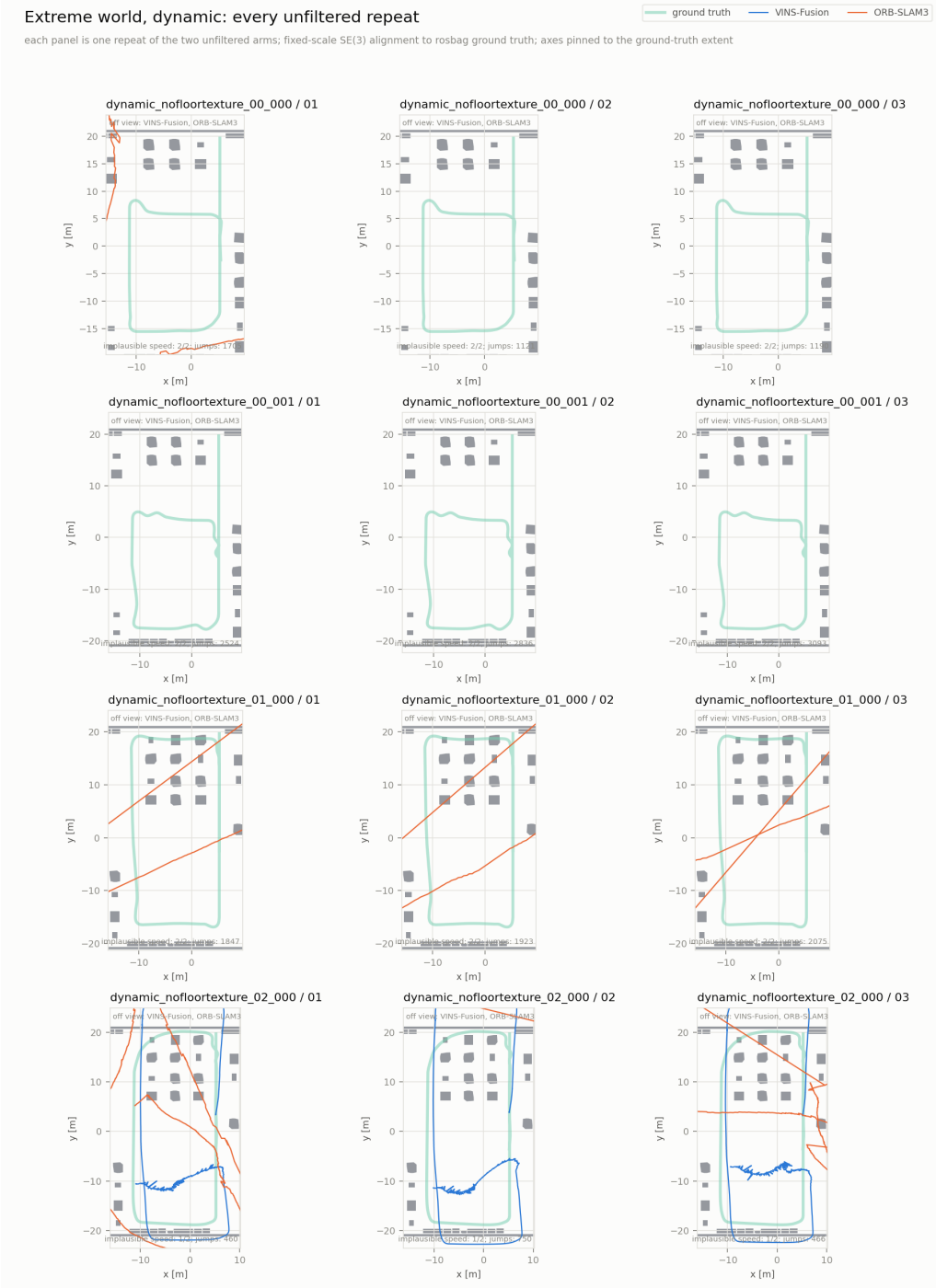


Figure 3.3: Extreme world, dynamic recordings: every repeat of the two unfiltered arms after rigid alignment, axes pinned to the ground-truth extent. Each panel names the systems that leave its frame. The VINS-Fusion trace of `dynamic_00_000` repeat 1 is the 15 October re-run.

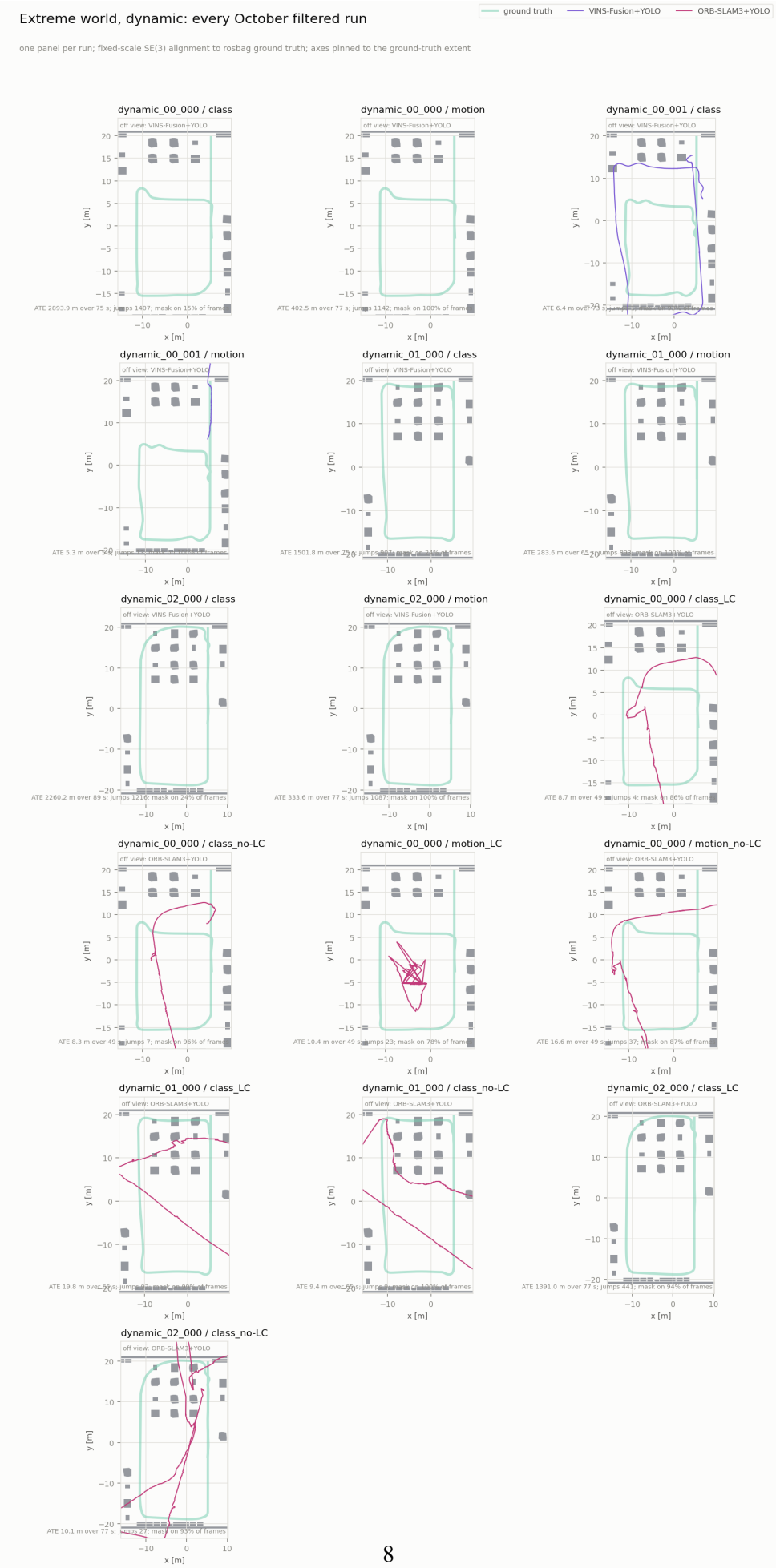


Figure 3.4: Extreme world, dynamic recordings: every October filtered run, one panel each, after rigid alignment with axes pinned to the ground-truth extent. Each footer gives ATE, discontinuities and the fraction of frames that

half the route (figure 3.8); on `dynamic_01_000` and `02_000` they cross at 20–36 s, part-way down the first aisles where the unfiltered runs had failed at 3–18 s, one of four in a heading change. Of the seventeen filtered runs with a working filter and an onset, thirteen crossed while the filter was still working.

The class mask leaves most of the image, the motion gate nearly all of it. The class mask leaves a median 88–94 % of the image area, with per-run minima down to 5 % on `dynamic_00_000` when a prop fills the view; the motion gate leaves 100 % at the median and 45–100 % at the minimum, because it rarely paints. The feature rows of figures 3.5 to 3.11 show the cost of masking as a percentage of each estimator's feature budget (ORB-SLAM3 keypoints after extraction against `ORBextractor.nFeatures`, VINS-Fusion tracked features against `max_cnt`), one figure per recording.

Table 3.6: Extreme world, unfiltered runs: onset of sustained error and its place on the route, per run, under the same definitions as table 3.7. One entry per run, in repeat order; "never" means the error stayed below the threshold for the whole run.

Recording	Estimator	LC	Onset [s] (where)	First jump [s]
`dynamic_00_000`	ORB-SLAM3	active	13 (straight 14 %); 14 (straight 15 %); 15 (straight 17 %)	6; 5; 3
	VINS-Fusion	—	18 (straight 22 %); 15 (straight 16 %); 32 (turn 35 %)	6; 4; 4
`dynamic_00_001`	ORB-SLAM3	active	16 (straight 21 %); 16 (straight 21 %); 16 (straight 21 %)	3; 3; 3
	VINS-Fusion	—	18 (straight 24 %); 16 (straight 22 %); 16 (straight 22 %)	1; 1; 1
`dynamic_01_000`	ORB-SLAM3	active	18 (straight 20 %); 6 (straight 5 %); 15 (straight 17 %)	2; 2; 2
	VINS-Fusion	—	16 (straight 17 %); 16 (straight 17 %); 16 (straight 17 %)	14; 14; 15
`dynamic_02_000`	ORB-SLAM3	active	3 (straight 2 %); 3 (straight 2 %); 3 (straight 2 %)	3; 3; 1
	VINS-Fusion	—	16 (straight 17 %); 17 (straight 18 %); 13 (straight 14 %)	2; 2; 2
`static_000`	ORB-SLAM3	active	49 (straight 57 %); 39 (straight 45 %); 36 (turn 42 %)	3; 3; 3
	VINS-Fusion	—	70 (turn 83 %); 70 (turn 83 %); 70 (turn 82 %)	—; —; —
`static_001`	ORB-SLAM3	active	never; 30 (straight 37 %); never	3; 3; 3
	VINS-Fusion	—	40 (straight 50 %); 38 (straight 47 %); 69 (straight 86 %)	—; —; —
`static_002`	ORB-SLAM3	active	never; never; never	3; 3; 3
	VINS-Fusion	—	never; never; never	—; —; —

Real-time and resources. Neither pipeline keeps up with the input at half replay rate, and the detector's cost lands on ORB-SLAM3's tracking thread (table 3.8). ORB-SLAM3 kept 61 % of received frames on the static recordings and 89 % on the dynamic ones; VINS-Fusion keeps every frame that reaches it and absorbs the load as latency. With the class mask ORB-SLAM3's tracking median rises from 63 to 113 ms on the dynamic recordings and its kept fraction falls to 67 %, at the same CPU. The VINS-Fusion+YOLO rows equal the VINS-Fusion rows, as they must: the filter did not activate in those runs.

Loop closure. The native ORB-SLAM3 back-end paths executed, but no correction benefit can be claimed. Baseline ORB-SLAM3 recorded six loop closures, two in each repeat of `static_002`, each with two global bundle adjustments; one baseline dynamic run and one ORB-SLAM3+YOLO run recorded a map merge. Without an event-aligned before/after counterfactual, and with the dynamic trajectories failing plausibility, these counters show only that the code ran.

Summary. In the static recordings the best of three repeats reaches 0.11–0.25 m for VINS-Fusion and 0.10–0.56 m for ORB-SLAM3, with medians up to 0.74 m. The dynamic numbers are withheld: even the best repeat is wrong by more than the distance travelled, and the environment is not physical. The filtering hypothesis is untested here: the two comparable pairs lie in an invalid testbed and the fourteen October runs are unpaired. Unfiltered, every dynamic run fails on the first aisle; a running mask delays the failure by tens of seconds, to the first corner on one recording, but does not prevent it.

Table 3.7: Extreme world, filtered runs: when the error rises, where, and how long the filter ran. "Onset" is the first time the start-anchored position error stays above 3× the estimator's static median for 1 s (threshold in the text); "where" gives the route progress at that moment and whether the robot was on a straight or in a heading change (ground-truth yaw rate above $10\,^\circ\,s^{-1}$). "Filter until" is the end of the initial stretch of 5 s windows with a mask on at least 98 % of frames; "unmasked" is the image area left after masking on matched frames, median (minimum). Times are seconds from the ground-truth start.

Recording	Estimator	Mask	LC	Run	Onset [s]	Where	First jump [s]	Filter until	Match	Unmasked [%]
dynamic_00_000	ORB-SLAM3	class	active	class_loopclosing_noatlas	41.9	heading change, 51 %	30.8	38 s	0.86	89 (5)
	ORB-SLAM3	class	disabled	class_noloopclosing_noatlas	41.7	heading change, 51 %	30.7	68 s	0.96	89 (5)
	ORB-SLAM3	class	active	rep 01	37.6	straight, 44 %	2.8	5 s	0.37	90 (53)
	ORB-SLAM3	class	active	rep 02	15.6	straight, 18 %	10.4	never	0.02	90 (85)
	ORB-SLAM3	class	active	rep 03	15.5	straight, 18 %	5.0	never	0.01	90 (85)
	ORB-SLAM3	motion	disabled	motion_loopclosing_noatlas	43.1	heading change, 53 %	30.8	38 s	0.78	100 (74)
	ORB-SLAM3	motion	disabled	motion_noloopclosing_noatlas	41.8	heading change, 51 %	31.0	never	0.87	100 (45)
	VINS-Fusion	class	—	class	18.0	straight, 22 %	3.7	never	0.15	91 (81)
	VINS-Fusion	class	—	rep 01	5.8	straight, 3 %	48.2	whole run	1.00	90 (50)
	VINS-Fusion	class	—	rep 02	7.2	straight, 5 %	4.3	never	0.00	—
	VINS-Fusion	class	—	rep 03	15.5	straight, 18 %	4.3	never	0.00	—
	VINS-Fusion	motion	—	motion	20.0	straight, 23 %	3.8	whole run	1.00	100 (84)
dynamic_00_001	ORB-SLAM3	class	active	rep 01	6.7	straight, 7 %	3.0	whole run	1.00	91 (31)
	ORB-SLAM3	class	active	rep 02	4.2	straight, 3 %	3.0	never	0.72	90 (31)
	ORB-SLAM3	class	active	rep 03	7.5	straight, 8 %	3.0	never	0.47	90 (31)
	VINS-Fusion	class	—	class	27.6	heading change, 37 %	3.3	60 s	0.93	92 (31)
	VINS-Fusion	class	—	rep 01	16.3	straight, 22 %	1.0	never	0.00	—
	VINS-Fusion	class	—	rep 02	16.3	straight, 22 %	1.0	never	0.00	—
	VINS-Fusion	class	—	rep 03	16.3	straight, 22 %	1.0	never	0.00	—
	VINS-Fusion	motion	—	motion	5.4	straight, 5 %	3.2	whole run	1.00	100 (100)
dynamic_01_000	ORB-SLAM3	class	active	class_loopclosing_noatlas	36.4	heading change, 42 %	21.9	never	0.99	92 (70)
	ORB-SLAM3	class	disabled	class_noloopclosing_noatlas	22.6	straight, 25 %	22.0	never	1.00	92 (70)
	ORB-SLAM3	class	active	rep 01	3.2	straight, 2 %	2.6	60 s	0.80	93 (64)
	ORB-SLAM3	class	active	rep 02	3.1	straight, 2 %	2.5	20 s	0.58	94 (72)
	ORB-SLAM3	class	active	rep 03	3.2	straight, 2 %	2.6	never	0.38	91 (74)
	VINS-Fusion	class	—	class	0.4	straight, 0 %	28.4	10 s	0.24	88 (73)
	VINS-Fusion	class	—	rep 01	15.6	straight, 17 %	15.1	never	0.00	—
	VINS-Fusion	class	—	rep 02	15.7	straight, 17 %	15.1	never	0.00	—
	VINS-Fusion	class	—	rep 03	15.6	straight, 17 %	15.0	never	0.00	—
	VINS-Fusion	motion	—	motion	27.1	heading change, 31 %	24.2	never	1.00	100 (75)
dynamic_02_000	ORB-SLAM3	class	active	class_loopclosing_noatlas	24.6	straight, 27 %	11.3	never	0.94	89 (63)
	ORB-SLAM3	class	disabled	class_noloopclosing_noatlas	19.9	straight, 21 %	11.2	44 s	0.93	88 (63)
	ORB-SLAM3	class	active	rep 01	3.3	straight, 2 %	2.7	never	0.31	93 (65)
	ORB-SLAM3	class	active	rep 02	3.3	straight, 2 %	2.6	10 s	0.34	94 (69)
	ORB-SLAM3	class	active	rep 03	3.2	straight, 2 %	2.6	10 s	0.37	92 (57)
	VINS-Fusion	class	—	class	15.9	straight, 17 %	15.1	10 s	0.24	89 (69)
	VINS-Fusion	class	—	rep 01	16.6	straight, 17 %	3.0	never	0.00	—
	VINS-Fusion	class	—	rep 02	13.2	straight, 14 %	1.9	never	0.00	—
	VINS-Fusion	class	—	rep 03	15.4	straight, 16 %	1.9	never	0.00	—
	VINS-Fusion	motion	—	motion	20.3	straight, 22 %	10.8	whole run	1.00	100 (64)

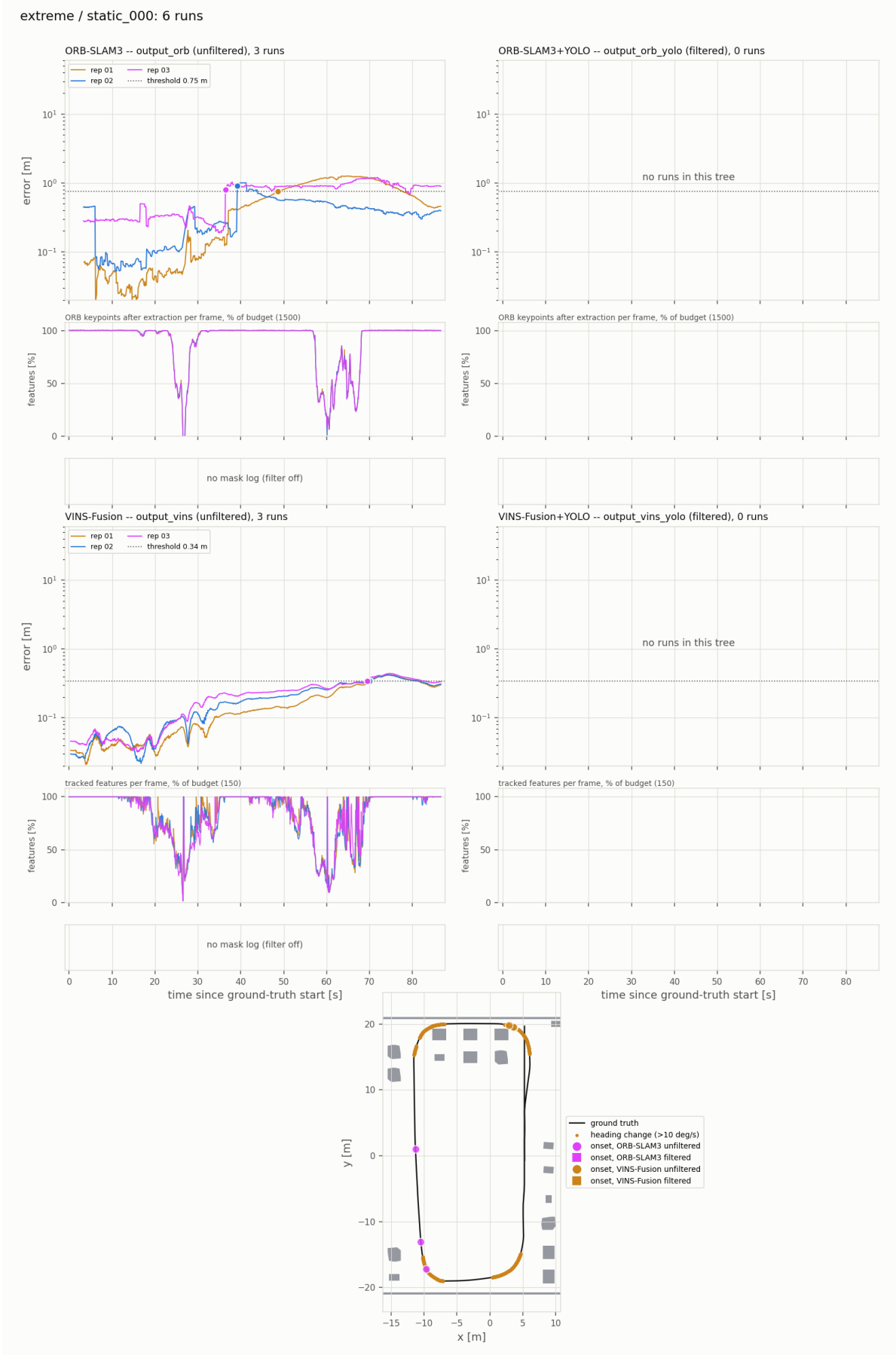


Figure 3.5: Extreme world, `static_000`, every run. Rows: ORB-SLAM3, VINS-Fusion; columns: unfiltered and filtered output trees; one hue per run (runs five to eight dashed). Each error panel is the start-anchored position error on a log scale with the estimator's threshold (dotted) and each run's onset (dot); beneath it the features per frame as a percentage of the estimator's budget; beneath each filtered cell the image area left after masking, with a gap wherever no mask was present. Bottom: the ground-truth route on the warehouse footprint, heading-change segments in amber, every onset placed on it (circles unfiltered, squares filtered).

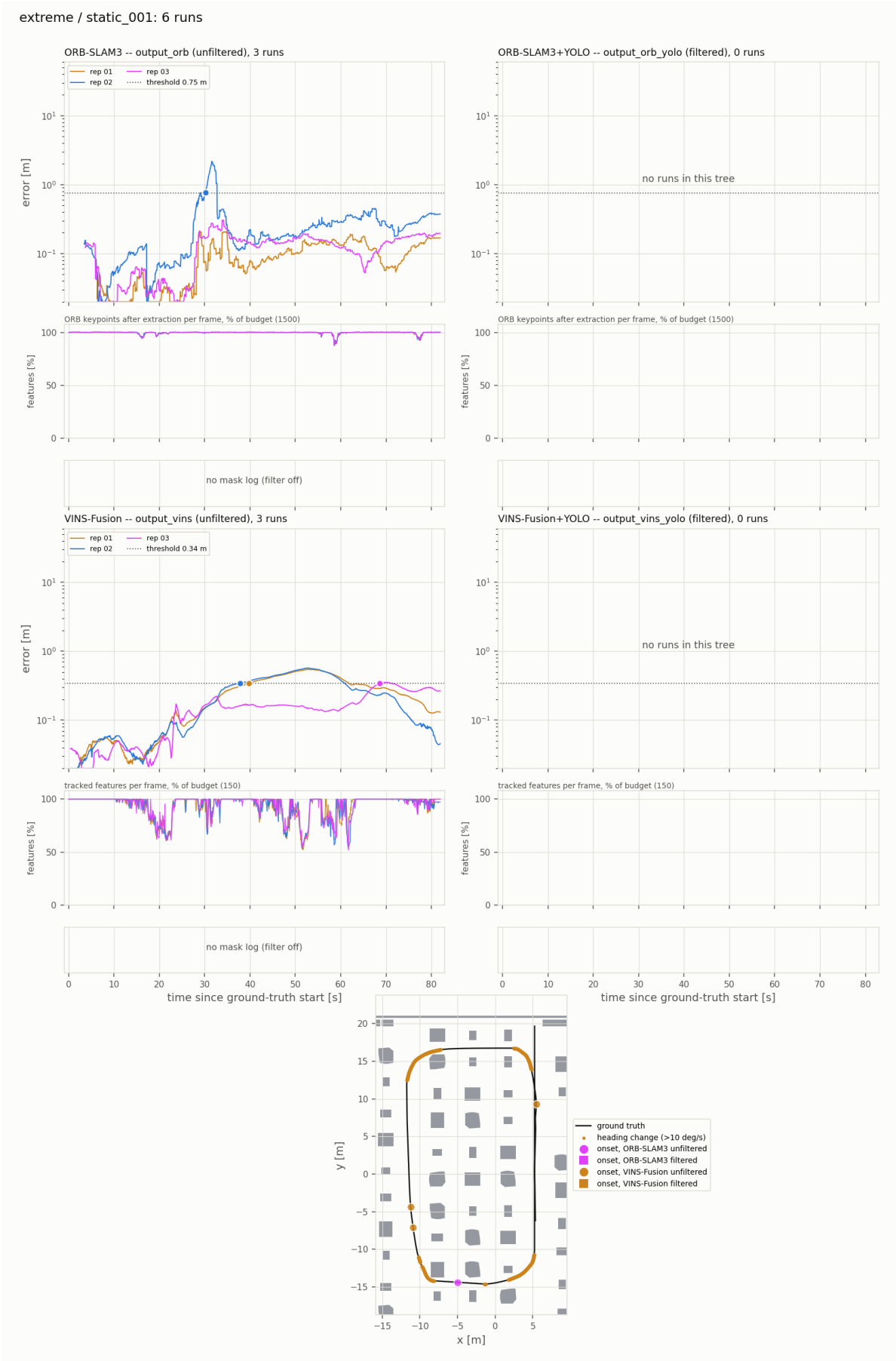


Figure 3.6: Extreme world, `static_001`, every run; panels as in figure 3.5.

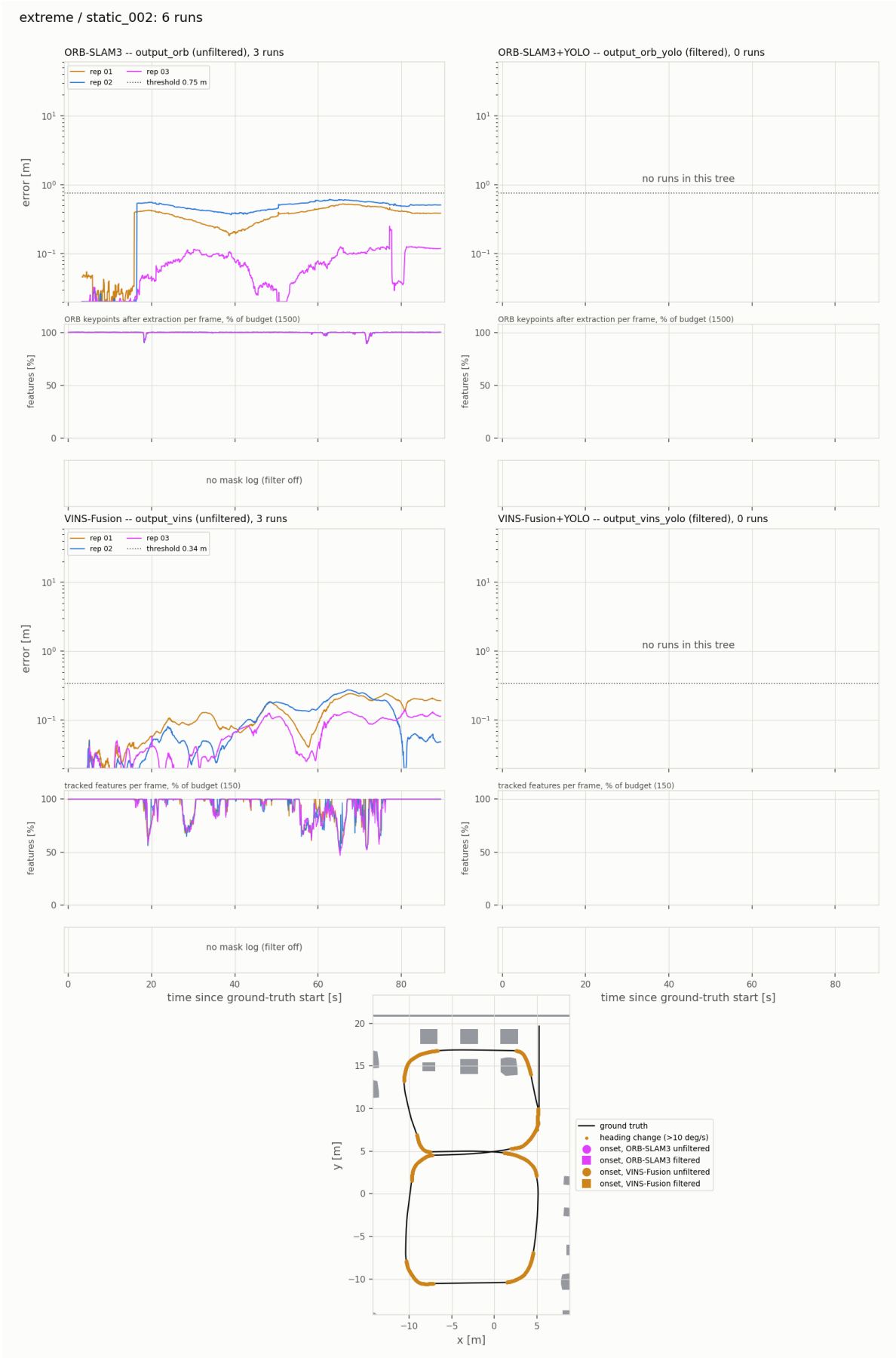


Figure 3.7: Extreme world, `static_002`, every run; panels as in figure 3.5.

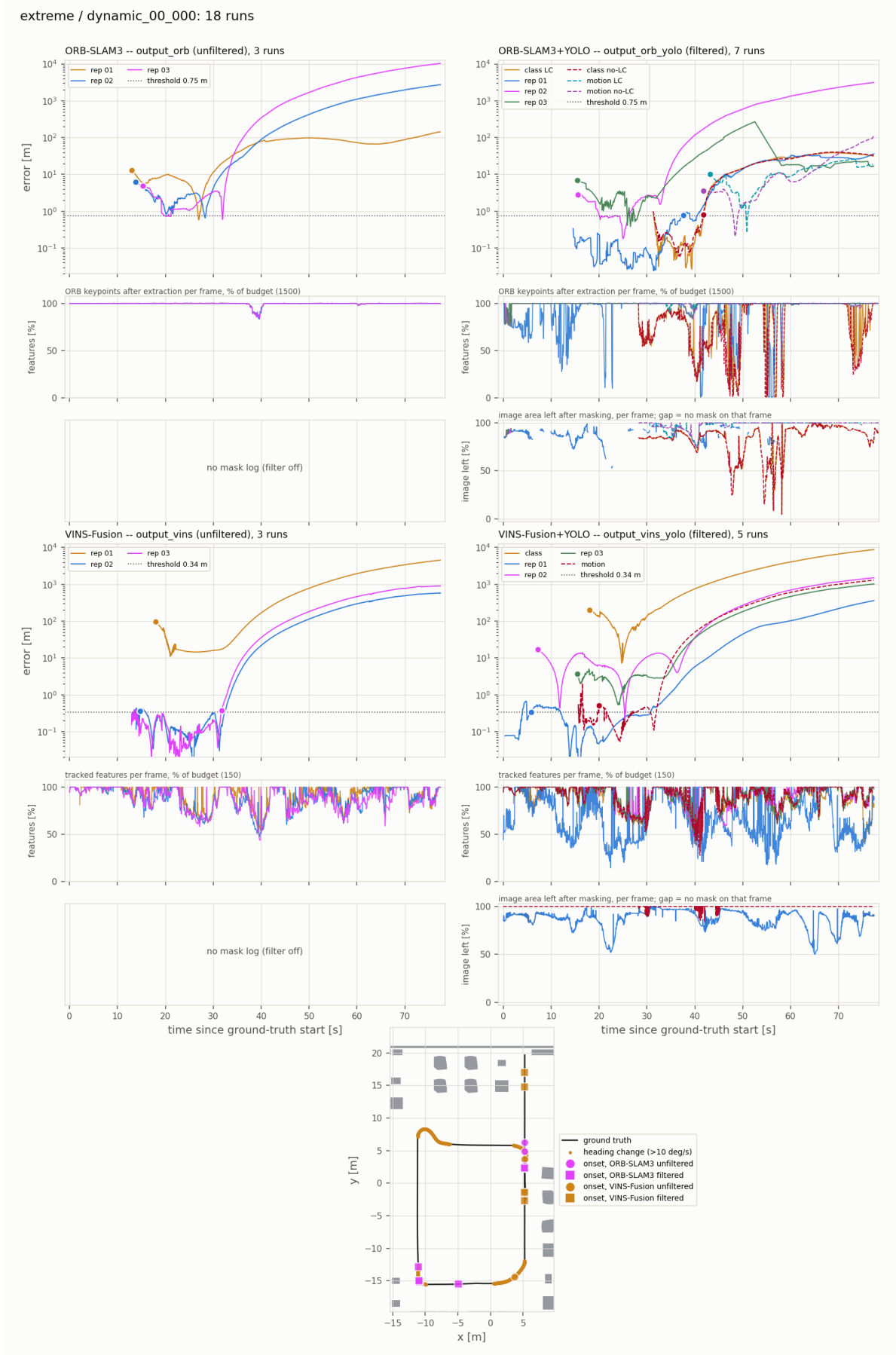


Figure 3.8: Extreme world, `dynamic_00_000`, every run; panels as in figure 3.5. The four October ORB-SLAM3 runs survive the first two aisles and cross in the south-west corner at 42–43 s; the September runs cross on the first aisle.

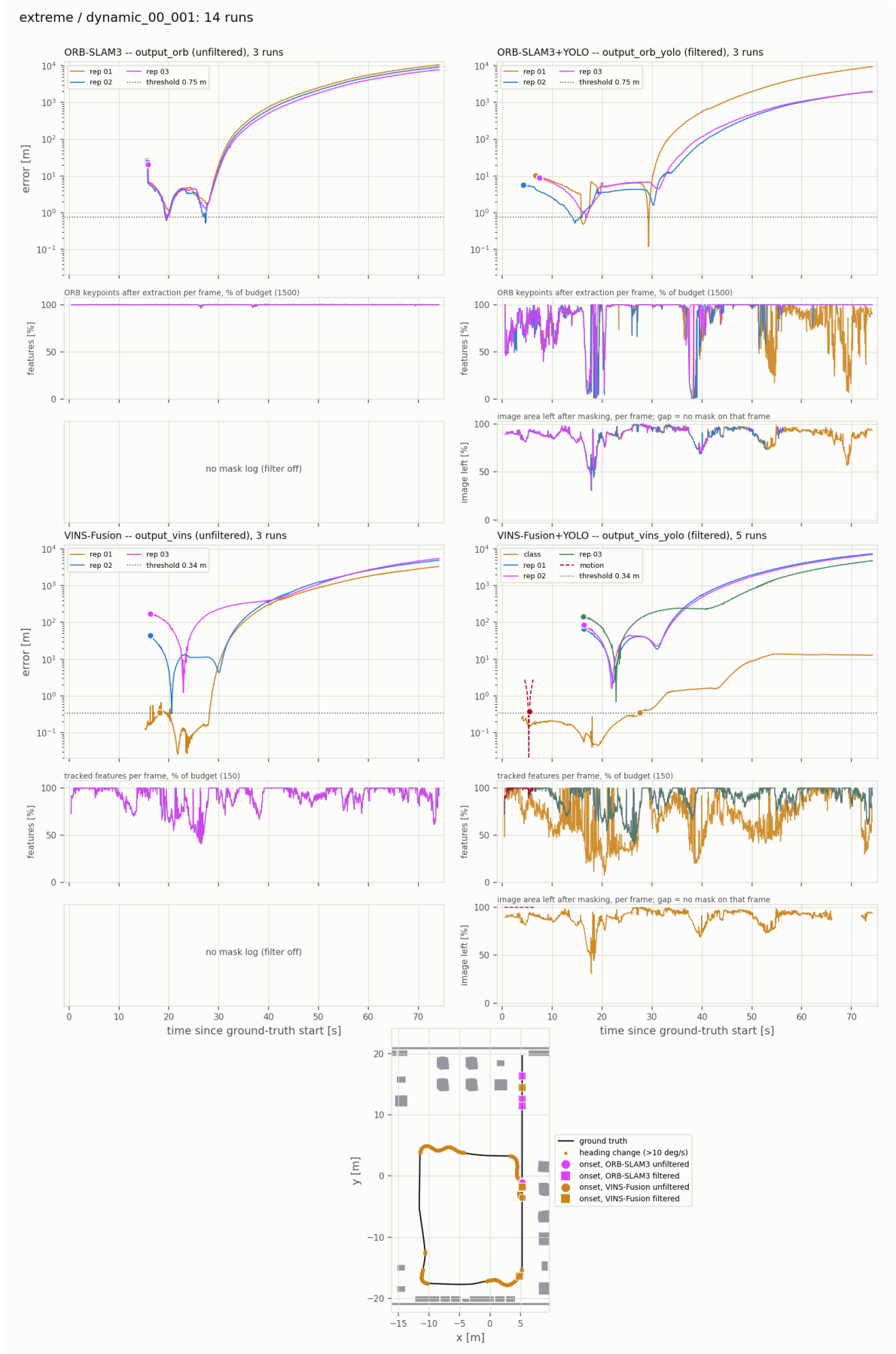


Figure 3.9: Extreme world, `dynamic_00_001`, every run; panels as in figure 3.5.

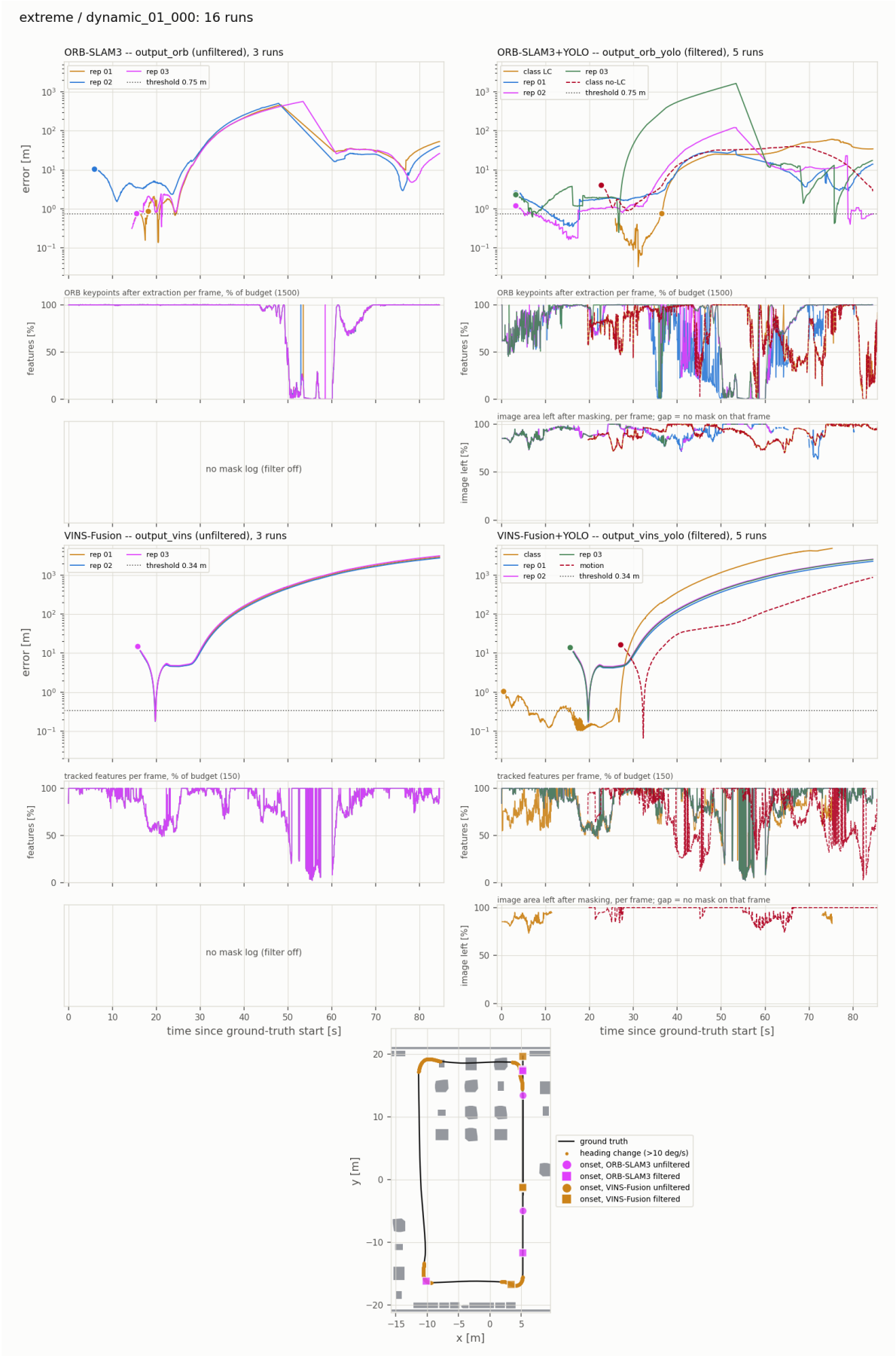


Figure 3.10: Extreme world, `dynamic_01_000`, every run; panels as in figure 3.5.

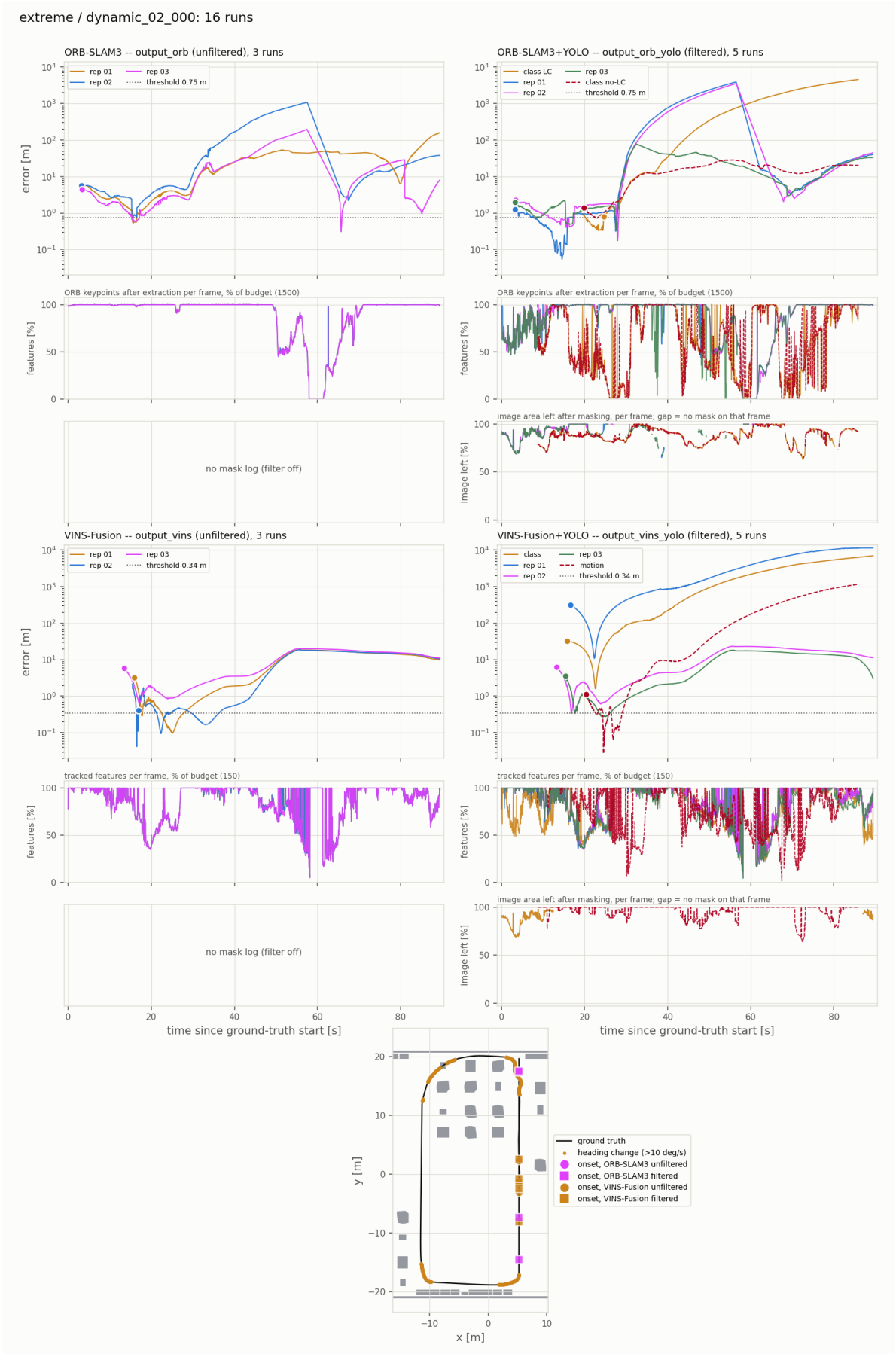


Figure 3.11: Extreme world, `dynamic_02_000`, every run; panels as in figure 3.5.

Table 3.8: Extreme world, real-time and resource measurements: mean ± SD over the three September repeats per recording (n), with a pooled row per scene. Front-end and back-end are each run's median module latency (ORB-SLAM3: tracking and local mapping; VINS-Fusion: feature tracking and back-end optimisation); throughput is processed frames per second of sensor time; "Kept" the processed fraction of received frames; CPU is process time over wall time, in percent of one core; RSS the peak resident set. Replay rate 0.5×. The VINS-Fusion+YOLO rows are the September runs, in which the filter did not activate except on `dynamic_00_000` repeat 1 (re-run 15 October).

Recording	Estimator	n	Front-end [ms]	Back-end [ms]	Throughput [Hz]	Kept [%]	CPU [%]	Peak RSS [Mi B]
`static_000`	VINS-Fusion	3	80.2 ± 4.9	95.5 ± 3.7	12.9 ± 0.7	100 ± 0	725 ± 8	1071 ± 31
	ORB-SLAM3	3	82.5 ± 6.6	411.0 ± 25.8	14.1 ± 0.6	60 ± 2	440 ± 4	1010 ± 13
`static_001`	VINS-Fusion	3	83.4 ± 4.9	103.0 ± 3.3	11.7 ± 0.6	100 ± 0	772 ± 9	1099 ± 6
	ORB-SLAM3	3	83.7 ± 2.3	424.2 ± 11.4	12.1 ± 0.3	51 ± 2	442 ± 1	1009 ± 4
`static_002`	VINS-Fusion	3	78.2 ± 5.5	99.2 ± 3.2	13.1 ± 0.8	100 ± 0	770 ± 11	1090 ± 12
	ORB-SLAM3	3	58.4 ± 5.4	327.6 ± 24.6	17.4 ± 1.4	72 ± 5	434 ± 7	1169 ± 21
all static	VINS-Fusion	9	80.6 ± 5.0	99.2 ± 4.4	12.5 ± 0.9	100 ± 0	756 ± 25	1087 ± 21
	ORB-SLAM3	9	74.8 ± 13.1	387.6 ± 49.1	14.5 ± 2.4	61 ± 9	438 ± 6	1062 ± 81
`dynamic_00_000`	VINS-Fusion	3	59.3 ± 15.9	77.0 ± 5.1	20.4 ± 6.5	100 ± 0	485 ± 62	1112 ± 58
	ORB-SLAM3	3	73.6 ± 2.8	202.2 ± 11.6	15.8 ± 0.7	67 ± 2	428 ± 2	1105 ± 11
	VINS-Fusion+YOLO	3	41.2 ± 1.6	48.3 ± 2.7	27.8 ± 1.1	100 ± 0	491 ± 8	1108 ± 66
	ORB-SLAM3+YOLO	3	121.9 ± 29.0	347.9 ± 66.2	9.9 ± 2.9	51 ± 10	430 ± 11	1000 ± 36
`dynamic_00_001`	VINS-Fusion	3	43.3 ± 3.5	72.4 ± 2.7	30.3 ± 0.0	100 ± 0	385 ± 22	411 ± 13
	ORB-SLAM3	3	60.4 ± 2.9	198.9 ± 9.9	27.4 ± 0.3	98 ± 0	377 ± 8	1292 ± 19
	VINS-Fusion+YOLO	3	40.4 ± 1.0	71.1 ± 0.4	30.3 ± 0.0	100 ± 0	364 ± 4	396 ± 4
	ORB-SLAM3+YOLO	3	104.8 ± 31.2	388.1 ± 103.1	19.6 ± 7.8	80 ± 18	381 ± 86	1147 ± 159
`dynamic_01_000`	VINS-Fusion	3	41.9 ± 1.0	54.5 ± 2.9	30.3 ± 0.0	100 ± 0	396 ± 10	468 ± 30
	ORB-SLAM3	3	60.5 ± 1.1	171.5 ± 19.0	25.0 ± 0.2	94 ± 1	360 ± 1	1220 ± 45
	VINS-Fusion+YOLO	3	39.8 ± 1.1	49.9 ± 7.9	30.3 ± 0.0	100 ± 0	323 ± 110	435 ± 23
	ORB-SLAM3+YOLO	3	97.6 ± 23.5	333.4 ± 90.4	15.7 ± 3.6	74 ± 10	408 ± 19	1044 ± 14
`dynamic_02_000`	VINS-Fusion	3	41.8 ± 0.9	36.7 ± 2.5	30.3 ± 0.0	100 ± 0	475 ± 15	551 ± 56
	ORB-SLAM3	3	58.3 ± 6.1	185.6 ± 10.2	25.7 ± 1.2	95 ± 2	376 ± 13	1287 ± 9
	VINS-Fusion+YOLO	3	53.0 ± 23.3	64.1 ± 28.0	26.7 ± 6.2	100 ± 0	451 ± 78	694 ± 306
	ORB-SLAM3+YOLO	3	125.9 ± 12.0	355.8 ± 9.7	12.7 ± 1.0	63 ± 2	425 ± 3	1028 ± 18
all dynamic	VINS-Fusion	12	46.6 ± 10.4	60.2 ± 16.9	27.8 ± 5.3	100 ± 0	435 ± 55	636 ± 294
	ORB-SLAM3	12	63.2 ± 7.1	189.6 ± 16.9	23.5 ± 4.8	89 ± 13	385 ± 28	1226 ± 82
	VINS-Fusion+YOLO	12	43.6 ± 11.5	58.3 ± 16.0	28.8 ± 3.2	100 ± 0	407 ± 91	658 ± 325
	ORB-SLAM3+YOLO	12	112.5 ± 24.6	356.3 ± 68.4	14.5 ± 5.4	67 ± 15	411 ± 43	1055 ± 91

4.3.1.2 Realistic world: physically simulated traffic

Objective. Measure what warehouse traffic, physically simulated, does to each pipeline, and whether a filter treatment can be tested at deployment rate.

Process. Every run replayed at rate 1.0 with the detector live, the deployment condition. Both estimators ran on all four recordings with the filter off (VINS-Fusion once; ORB-SLAM3 with its loop closer active and disabled), and on the two dynamic recordings with the class mask and the motion gate; eight ORB-SLAM3 runs additionally loaded a prior atlas. There is one run per condition, so no within-condition spread exists.

Datasets. `realistic_000` and `_001` static, `_002` and `_003` dynamic with ten and twelve props moving during the recording, in the U and I layouts respectively. The U-layout recordings do not share a route.

Reporting rules. Every run is reported on its own; accuracy, latency and resources per method and filter separately. Alignment and RPE as in the extreme world. The loop-closer state is read from `loop_closing.csv`, not from the run name. A filtered run is paired with the baseline of the same estimator and loop-closer state, and the pair is admissible only under three gates: an active filter (`detection_match_rate` ≥ 0.98), processed frames and duration within 5 % of the baseline, and both runs on the same node binary (built 16 October 2026). Prior-atlas runs are relocalisation experiments and are listed but not paired. Ground truth is the bag's `/ground_truth/odometry`, moved from the model origin to the IMU point both estimators report (lever arm 0.338 m forward, 0.192 m up, from the robot model) before alignment; the figures draw that moved path. Sources: `output/compare/simulation/exp-realistic-ab/`, `output/compare/simulation/exp3-timelines/`.

Results. With physically simulated traffic neither front end loses tracking: across both dynamic recordings ORB-SLAM3 holds 0.34–0.70 m aligned ATE against 0.29–0.59 m on the static twins (table 3.9, figures 3.12 and 3.13). In the extreme world the same estimators produced hundreds of metres of error per run.

Traffic degrades VINS-Fusion on one of two layouts. On the U layout, `realistic_002`, it reads 1.85 m against 0.32 m on the static twin, with a jump-free RPE of 0.21 m against 0.07–0.09 m everywhere else: a continuous degradation with no jump, which is what a tracker following features on moving objects looks like.

[Confound] Traffic is not separated from route: the U-layout recordings do not share a route, and there is one run per cell.

Loop closure changes nothing attributable to it. The closer fired zero to two times per run, and the two ORB-SLAM3 arms differ by 0.05–0.30 m in either direction.

ORB-SLAM3's rotational RMSE of 13–17° is one event. On every run the median rotational RPE at 1 s is 0.06–0.16° and 99 % of pairs lie under 5°; the remainder span a single orientation step of about 90° at 2.5–2.7 s, where inertial initialisation re-aligns the world frame to gravity. The jump detector is translational, so the jump-free rotational statistic does not exclude it; the median is the number to read.

No run is at input rate. At replay rate 1.0 the estimators processed 58–82 % of each bag's frames, ORB-SLAM3 by dropping at its queue, VINS-Fusion by losing frames before its own counter sees them.

Filtered runs: no admissible pair. Of twenty filtered runs, the three gates leave none (table 3.10). Eight loaded a prior atlas and are excluded as relocalisation runs. All four VINS-Fusion runs carried a current

Table 3.9: Unfiltered arms on the four realistic-warehouse recordings, 21 October 2026, replay rate 1.0. Aligned ATE is RMSE after a rigid $SE(3)$ fit with scale fixed; RPE is translational, at $\Delta = 1\,\mathrm{s}$ over pairs that do not span a jump; "Jumps" counts steps above $10\,\mathrm{m\,s^{-1}}$; "Kept" is the fraction of the bag's stereo pairs the estimator processed (frames lost before the node's own queue count as lost). Loop-closer state is read from `loop_closing.csv`, not from the run name. Static and dynamic rows of one layout share a world geometry and differ in traffic only.

Recording	Layout	Scene	Estimator	ATE [m]	Rot. [°]	$\mathrm{RPE}_{1\,\mathrm{s}}$ [m]	Jumps	Kept	Resets
`realistic_000`	U	static	ORB-SLAM3	0.591	13.15	0.113	2	0.60	0
			ORB-SLAM3 (no LC)	0.294	13.35	0.132	1	0.69	0
			VINS-Fusion	0.324	1.06	0.081	0	0.58	0
`realistic_001`	I	static	ORB-SLAM3	0.350	16.36	0.148	0	0.58	0
			ORB-SLAM3 (no LC)	0.492	16.40	0.102	2	0.59	0
			VINS-Fusion	0.607	3.94	0.089	0	0.67	0
`realistic_002`	U	dynamic	ORB-SLAM3	0.704	16.35	0.095	2	0.67	0
			ORB-SLAM3 (no LC)	0.575	17.19	0.092	2	0.69	0
			VINS-Fusion	1.845	3.62	0.211	0	0.73	0
`realistic_003`	I	dynamic	ORB-SLAM3	0.337	13.87	0.081	2	0.81	0
			ORB-SLAM3 (no LC)	0.387	14.21	0.072	1	0.73	0
			VINS-Fusion	0.243	1.47	0.069	0	0.82	0

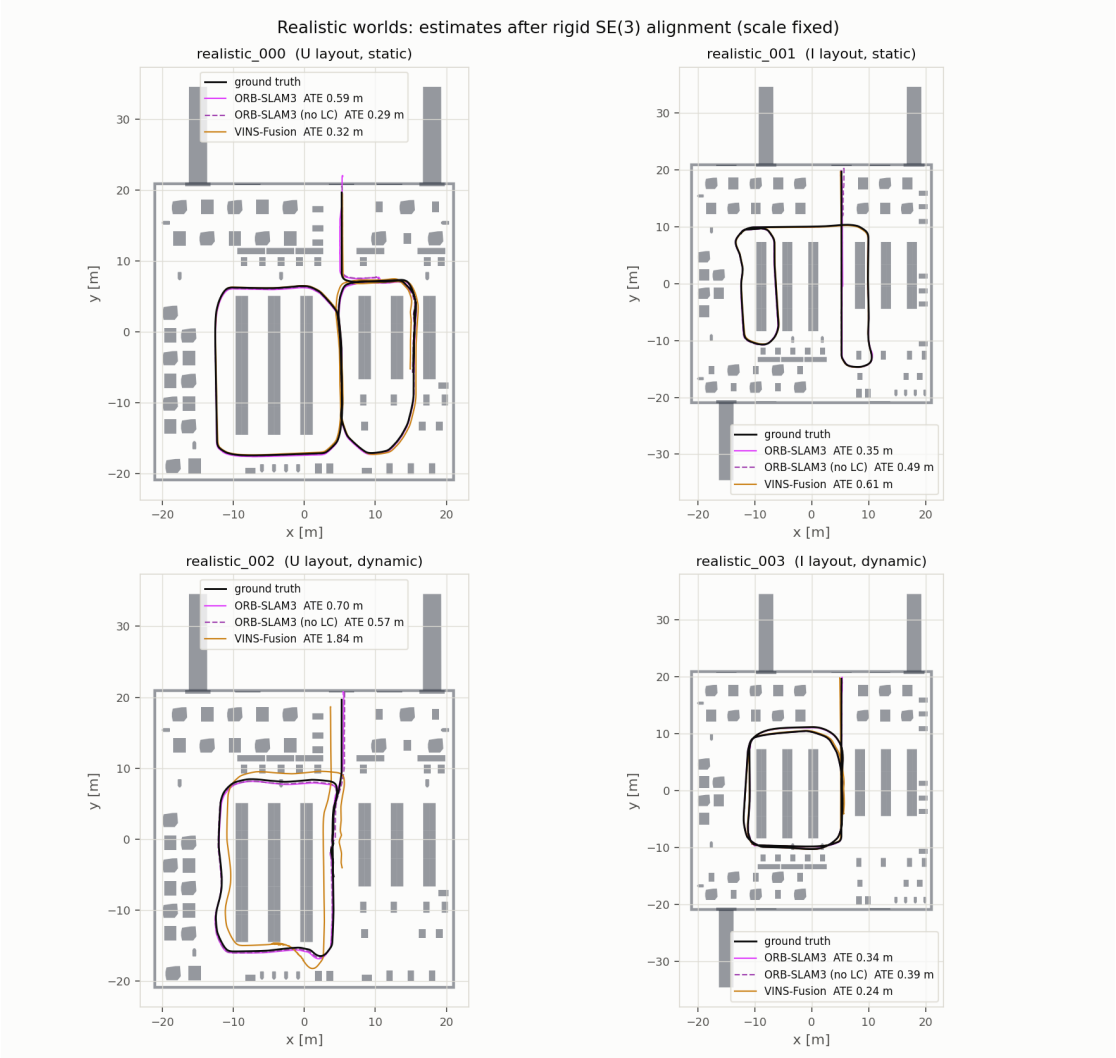


Figure 3.12: Unfiltered estimates on the four realistic-warehouse recordings after rigid $SE(3)$ alignment with scale fixed, on the warehouse footprint; black is ground truth. Top: the static twins; bottom: the same layouts with traffic. Dashed purple is ORB-SLAM3 with its loop closer disabled. The only visible departure is VINS-Fusion on `realistic_002`.

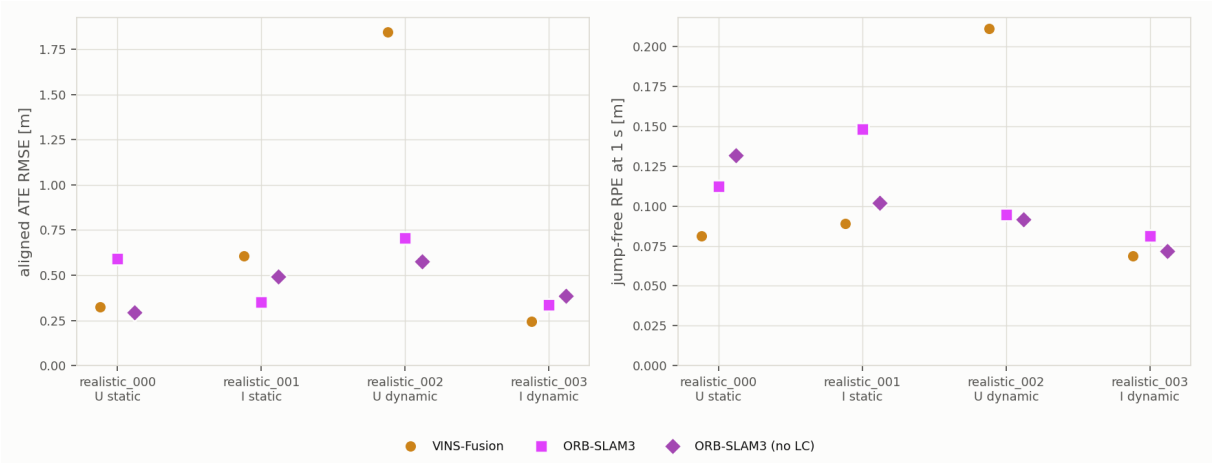


Figure 3.13: Aligned ATE (left) and jump-free translational RPE at 1 s (right) of the unfiltered arms, one marker per run.

mask on 18–39 % of frames (`filter_status degraded`): the live detector could not keep up at rate 1.0, so the filter did not run. Three ORB-SLAM3 runs had an active filter; two processed 15–33 % fewer frames than their baseline, and the third, the class mask with loop closure on `realistic_002`, is input-matched (4.7 %) but was recorded on 14 October with a node binary older than its baseline's. That pair reads 1.47 m filtered against 0.70 m unfiltered, four jumps against two; a number, not an effect.

Two patterns in the inadmissible runs are hypotheses for the repeat. The class-masked ORB-SLAM3 runs on the I layout carry 11–45 jumps where their baselines carry one or two, which would mean the mask removes features the tracker needs on the long rack faces. On that layout every run with the mask present on at least 97 % of frames processed 15–50 % fewer frames than its baseline, which is the detector's cost and, under the deployment definition, part of the result.

Table 3.10: Every filtered run on the dynamic realistic recordings, with the three admissibility gates of section 4.3.1.2. "Match" is `detection_match_rate`, the fraction of frames that carried a current detector mask; below 0.98 the filter did not run. "Input" is the processed-frame count relative to the paired baseline. "Binary" says whether the run and its baseline share the node build of 16 October 2026. A run passes only when all three hold; ATE and RPE are listed for every run but may be read as a treatment effect only on a passing row. Runs that loaded a prior atlas are outside the A/B and carry no baseline.

Recording	Estimator	Mask	LC	Atlas	Match	Status	Input	Binary	ATE [m]	RPE_{1s} [m]	Jumps	Verdict
`realistic_002`	ORB-SLAM3	class	active	no	0.997	—	+5 %	older	1.473	0.206	4	binary
	ORB-SLAM3	class	active	yes	0.772	—	—	older	12.334	0.329	2	atlas, not paired
	ORB-SLAM3	motion	active	no	0.010	—	-53 %	older	0.313	0.126	1	filter inactive, input, binary
	ORB-SLAM3	motion	active	yes	0.728	—	—	older	9.825	0.145	6	atlas, not paired
	ORB-SLAM3	class	disabled	no	0.250	—	-35 %	older	1.273	0.080	2	filter inactive, input, binary
	ORB-SLAM3	class	disabled	yes	0.998	—	—	older	1.806	0.168	26	atlas, not paired
	ORB-SLAM3	motion	disabled	no	0.127	—	-44 %	older	0.683	0.083	2	filter inactive, input, binary
	ORB-SLAM3	motion	disabled	yes	0.795	—	—	older	1.248	0.358	10	atlas, not paired
	VINS-Fusion	class	—	no	0.207	degraded	-23 %	same	2.006	0.226	4	filter inactive, input
	VINS-Fusion	motion	—	no	0.394	degraded	-42 %	same	0.558	0.117	0	filter inactive, input
`realistic_003`	ORB-SLAM3	class	active	no	0.998	—	-15 %	older	2.022	0.153	11	input, binary
	ORB-SLAM3	class	active	yes	0.879	—	—	older	2.041	0.293	17	atlas, not paired
	ORB-SLAM3	motion	active	no	0.977	degraded	-50 %	same	10.369	1.170	10	filter inactive, input
	ORB-SLAM3	motion	active	yes	0.585	—	—	older	2.701	0.647	8	atlas, not paired
	ORB-SLAM3	motion	active	yes	0.782	—	—	older	1.335	0.477	10	atlas, not paired
	ORB-SLAM3	class	disabled	no	0.619	—	-32 %	older	12.143	1.348	45	filter inactive, input, binary
	ORB-SLAM3	class	disabled	yes	0.993	—	—	older	3.339	0.141	29	atlas, not paired
	ORB-SLAM3	motion	disabled	no	0.989	ok	-33 %	same	1.105	0.081	3	input
	VINS-Fusion	class	—	no	0.281	degraded	-48 %	same	0.548	0.084	0	filter inactive, input
	VINS-Fusion	motion	—	no	0.177	degraded	-38 %	same	0.279	0.071	0	filter inactive, input

Resources. The class mask costs ORB-SLAM3 almost nothing; the motion gate and the prior atlas halve its throughput; the live detector starves VINS-Fusion (table 3.11). Unfiltered, both pipelines run at 18–25 Hz, VINS-Fusion on about seven cores to ORB-SLAM3's four and with half the memory. ORB-

SLAM3 with the class mask and loop closure keeps 69–70 % of frames at 54–60 ms tracking, whereas the motion gate and every atlas run roughly double the tracking median and keep 22–50 %. VINS-Fusion filtered keeps 43–56 % of frames, which is the mechanism behind its low mask rates.

Table 3.11: Realistic world, real-time and resource measurements, one row per run (replay rate 1.0, live detector). Columns as in table 3.8; "Kept" here is the processed fraction of the bag's frames, so frames lost before the node's own queue count as lost. Accuracy for the same runs is in tables 3.9 and 3.10.

Recording	Estimator	Mask	LC	Atlas	Front-end [ms]	Back-end [ms]	Throughput [Hz]	Kept	CPU [%]	Peak RSS [Mi B]
realistic_000	ORB-SLAM3	off	active	no	57.3	313.9	18.2	60 %	422	1167
	ORB-SLAM3	off	disabled	no	49.4	260.6	21.0	69 %	407	1205
	VINS-Fusion	off	—	no	50.6	63.6	17.7	58 %	780	480
realistic_001	ORB-SLAM3	off	active	no	63.7	321.4	17.5	58 %	424	1101
	ORB-SLAM3	off	disabled	no	59.2	311.1	18.5	59 %	415	1094
	VINS-Fusion	off	—	no	47.8	62.2	20.2	67 %	734	478
realistic_002	ORB-SLAM3	off	active	no	59.8	308.7	20.2	67 %	416	1182
	ORB-SLAM3	off	disabled	no	58.4	300.6	20.8	69 %	404	1060
	VINS-Fusion	off	—	no	47.5	61.1	22.2	73 %	721	461
	ORB-SLAM3+YOLO	class	active	no	54.3	233.0	21.2	70 %	404	1170
	ORB-SLAM3+YOLO	class	active	yes	147.7	530.1	8.4	28 %	467	1153
	ORB-SLAM3+YOLO	motion	active	no	129.6	570.7	9.6	32 %	454	1083
	ORB-SLAM3+YOLO	motion	active	yes	160.7	644.3	7.8	26 %	462	1121
	ORB-SLAM3+YOLO	class	disabled	no	97.0	426.5	13.4	44 %	429	1028
	ORB-SLAM3+YOLO	class	disabled	yes	58.9	261.2	18.8	62 %	385	1204
	ORB-SLAM3+YOLO	motion	disabled	no	106.6	511.0	11.6	38 %	435	1008
	ORB-SLAM3+YOLO	motion	disabled	yes	135.6	545.9	8.9	29 %	443	1136
	VINS-Fusion+YOLO	class	—	no	63.6	85.3	17.1	56 %	718	485
	VINS-Fusion+YOLO	motion	—	no	86.3	103.5	12.9	43 %	701	489
realistic_003	ORB-SLAM3	off	active	no	54.2	292.5	24.8	81 %	398	1190
	ORB-SLAM3	off	disabled	no	62.0	323.5	22.0	73 %	402	1182
	VINS-Fusion	off	—	no	48.2	63.7	25.0	82 %	709	494
	ORB-SLAM3+YOLO	class	active	no	59.6	253.6	20.9	69 %	409	1197
	ORB-SLAM3+YOLO	class	active	yes	126.6	480.0	10.6	35 %	459	1243
	ORB-SLAM3+YOLO	motion	active	no	114.2	478.0	12.3	41 %	434	1145
	ORB-SLAM3+YOLO	motion	active	yes	214.9	745.0	6.7	22 %	446	1164
	ORB-SLAM3+YOLO	motion	active	yes	164.9	654.0	8.7	29 %	442	1223
	ORB-SLAM3+YOLO	class	disabled	no	92.0	271.4	15.1	50 %	424	1220
	ORB-SLAM3+YOLO	class	disabled	yes	77.8	321.4	17.1	56 %	411	1307
	ORB-SLAM3+YOLO	motion	disabled	no	93.3	450.1	15.0	49 %	415	1133
	VINS-Fusion+YOLO	class	—	no	94.3	107.3	13.0	43 %	673	471
	VINS-Fusion+YOLO	motion	—	no	77.8	100.7	15.6	51 %	736	486

Analysis: when, where, and how long the filter ran. Unfiltered, only one of six runs on the dynamic recordings ever crosses its threshold (0.83 m VINS-Fusion, 1.41 m ORB-SLAM3): VINS-Fusion on realistic_002 at 30 s, in the heading change at 39 % of the route (table 3.12, figure 3.16). Neither ORB-SLAM3 arm crosses on either dynamic recording.

Filtered, the class-masked ORB-SLAM3 runs fail at or just after the first heading change following the long rack aisle, with the mask present (table 3.13). Fourteen of twenty filtered runs cross, eight in a heading change; on realistic_003 all four class-masked ORB-SLAM3 runs cross, two in the first corner at 33–34 s (28–29 % of the route) and two on the straights after it at 39 and 87 s (figure 3.17), including the run whose mask was present on every frame. The six that never cross are motion-gated ORB-SLAM3 runs and VINS-Fusion runs whose masks were present on 1–39 % of frames, with one exception: ORB-SLAM3's motion gate without loop closure on realistic_003, mask present on 99 % of frames, no onset. With one run per cell this is where to look first in the repeat, not an effect.

The class mask leaves a median 90–93 % of the image (minima 49–85 %); the motion gate leaves 100 % (minima 62–100 %). The features-left rows of figures 3.14 to 3.17 give the per-frame cost against each estimator's budget.

Summary. With physically simulated traffic neither front end loses tracking: ORB-SLAM3 stays near its static range and VINS-Fusion degrades on one of two layouts, to 1.85 m, continuously and without a jump, from one run per condition with traffic not separated from route. The filter had no admissible test: the VINS-Fusion filter did not run at deployment rate, and every ORB-SLAM3 pair fails on input match or node binary. Where the class-masked ORB-SLAM3 runs fail, it is at or just after the first heading change following the long rack aisle, with the mask present; the unfiltered runs do not fail there.

Table 3.12: Realistic world, unfiltered runs: onset of sustained error and its place on the route, per run, under the same definitions as table 3.13. One entry per run, in repeat order; "never" means the error stayed below the threshold for the whole run.

Recording	Estimator	LC	Onset [s] (where)	First jump [s]
realistic_000	ORB-SLAM3	active	never	6
	ORB-SLAM3	disabled	never	6
	VINS-Fusion	—	110 (straight 90 %)	—
realistic_001	ORB-SLAM3	active	never	—
	ORB-SLAM3	disabled	never	5
	VINS-Fusion	—	never	—
realistic_002	ORB-SLAM3	active	never	3
	ORB-SLAM3	disabled	never	3
	VINS-Fusion	—	30 (turn 39 %)	—
realistic_003	ORB-SLAM3	active	never	4
	ORB-SLAM3	disabled	never	3
	VINS-Fusion	—	never	—

Table 3.13: Realistic world, filtered runs: when the error rises, where, and how long the filter ran. "Onset" is the first time the start-anchored position error stays above 3× the estimator's static median for 1 s (threshold in the text); "where" gives the route progress at that moment and whether the robot was on a straight or in a heading change (ground-truth yaw rate above $10\,^\circ\,s^{-1}$). "Filter until" is the end of the initial stretch of 5 s windows with a mask on at least 98 % of frames; "unmasked" is the image area left after masking on matched frames, median (minimum). Times are seconds from the ground-truth start.

Recording	Estimator	Mask	LC	Run	Onset [s]	Where	First jump [s]	Filter until	Match	Unmasked [%]
realistic_002	ORB-SLAM3	class	active	class_loopclosing_atlas (atlas)	17.4	straight, 22 %	17.4	never	0.77	92 (51)
	ORB-SLAM3	class	active	class_loopclosing_noatlas	30.7	heading change, 40 %	4.0	65 s	1.00	91 (49)
	ORB-SLAM3	class	disabled	class_noloopclosing_atlas (atlas)	27.4	heading change, 36 %	2.7	whole run	1.00	91 (50)
	ORB-SLAM3	class	disabled	class_noloopclosing_noatlas	16.8	straight, 21 %	2.9	never	0.25	90 (63)
	ORB-SLAM3	motion	active	motion_loopclosing_atlas (atlas)	3.5	straight, 2 %	2.9	never	0.73	100 (74)
	ORB-SLAM3	motion	active	motion_loopclosing_noatlas	never	—	3.0	never	0.01	100 (100)
	ORB-SLAM3	motion	disabled	motion_noloopclosing_atlas (atlas)	28.4	heading change, 37 %	2.8	never	0.79	100 (62)
	ORB-SLAM3	motion	disabled	motion_noloopclosing_noatlas	never	—	3.0	never	0.13	100 (100)
	VINS-Fusion	class	—	class	30.2	heading change, 39 %	33.1	never	0.21	90 (85)
	VINS-Fusion	motion	—	motion	never	—	—	never	0.39	100 (96)
realistic_003	ORB-SLAM3	class	active	class_loopclosing_atlas (atlas)	33.2	heading change, 28 %	2.6	never	0.88	91 (58)
	ORB-SLAM3	class	active	class_loopclosing_noatlas	38.9	straight, 34 %	2.6	never	1.00	91 (58)
	ORB-SLAM3	class	disabled	class_noloopclosing_atlas (atlas)	87.4	straight, 79 %	2.7	never	0.99	91 (58)
	ORB-SLAM3	class	disabled	class_noloopclosing_noatlas	33.9	heading change, 29 %	2.7	20 s	0.62	91 (58)
	ORB-SLAM3	motion	active	motion_loopclosing_atlas (atlas)	22.9	heading change, 19 %	3.0	never	0.59	100 (90)
	ORB-SLAM3	motion	active	motion_loopclosing_noatlas	36.3	straight, 31 %	2.9	never	0.98	100 (68)
	ORB-SLAM3	motion	active	motion_noloopclosing_atlas (atlas)	33.2	heading change, 28 %	2.9	never	0.78	100 (71)
	ORB-SLAM3	motion	disabled	motion_noloopclosing_noatlas	never	—	4.1	never	0.99	100 (68)
	VINS-Fusion	class	—	class	never	—	—	never	0.28	91 (65)
	VINS-Fusion	motion	—	motion	never	—	—	5 s	0.18	100 (100)

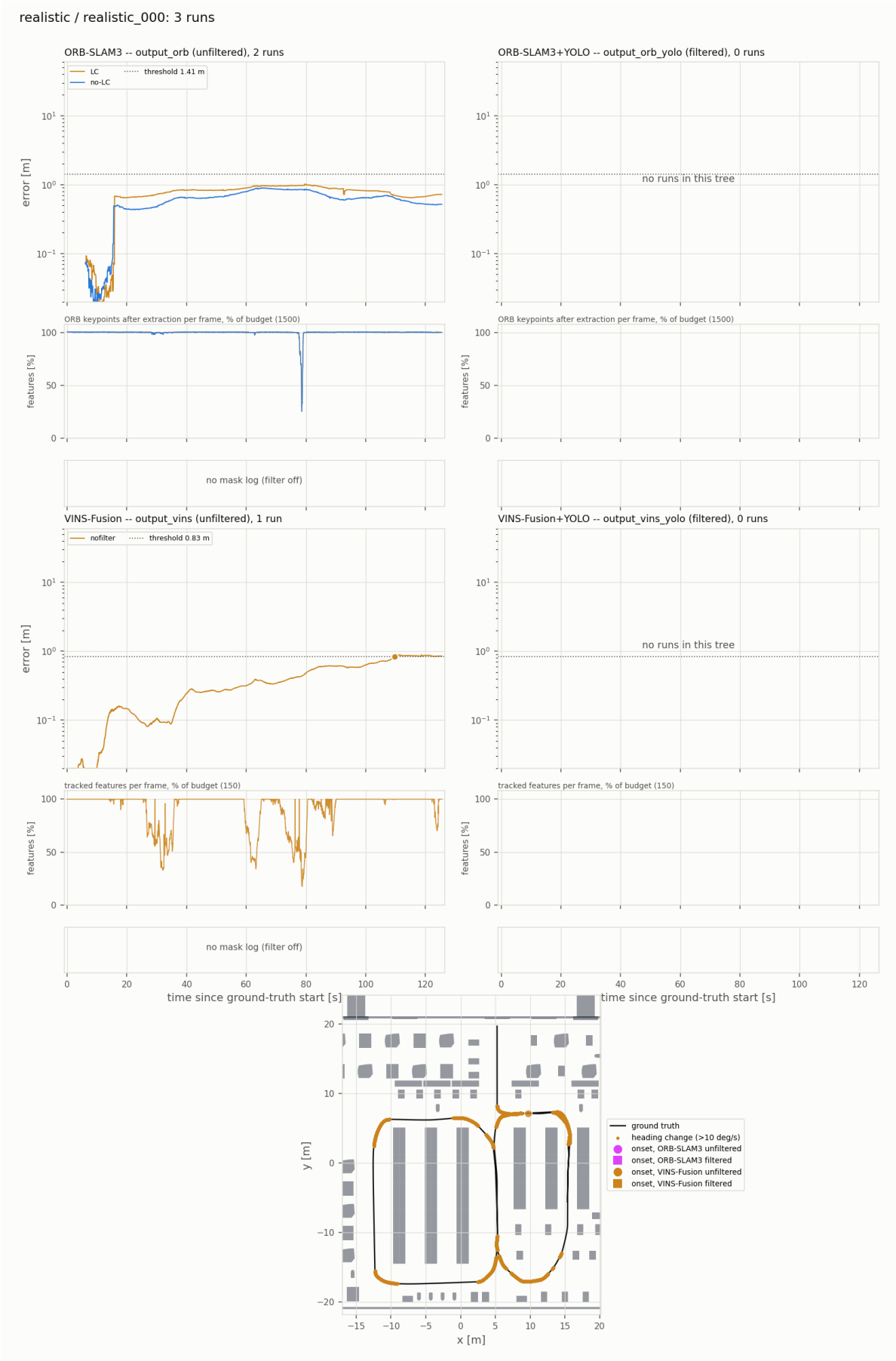


Figure 3.14: Realistic world, `realistic_000`, every run; panels as in figure 3.5.

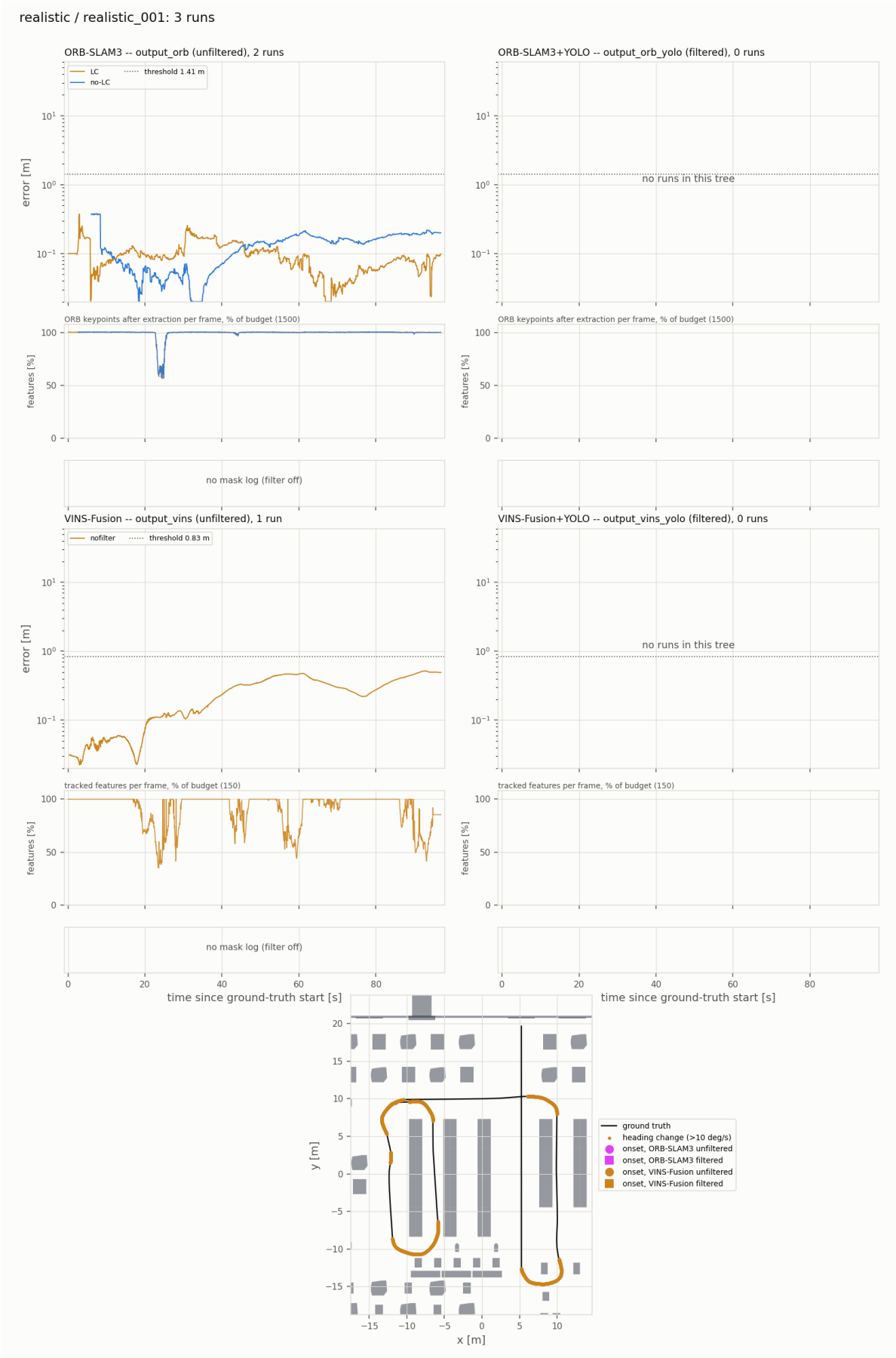


Figure 3.15: Realistic world, `realistic_001`, every run; panels as in figure 3.5.

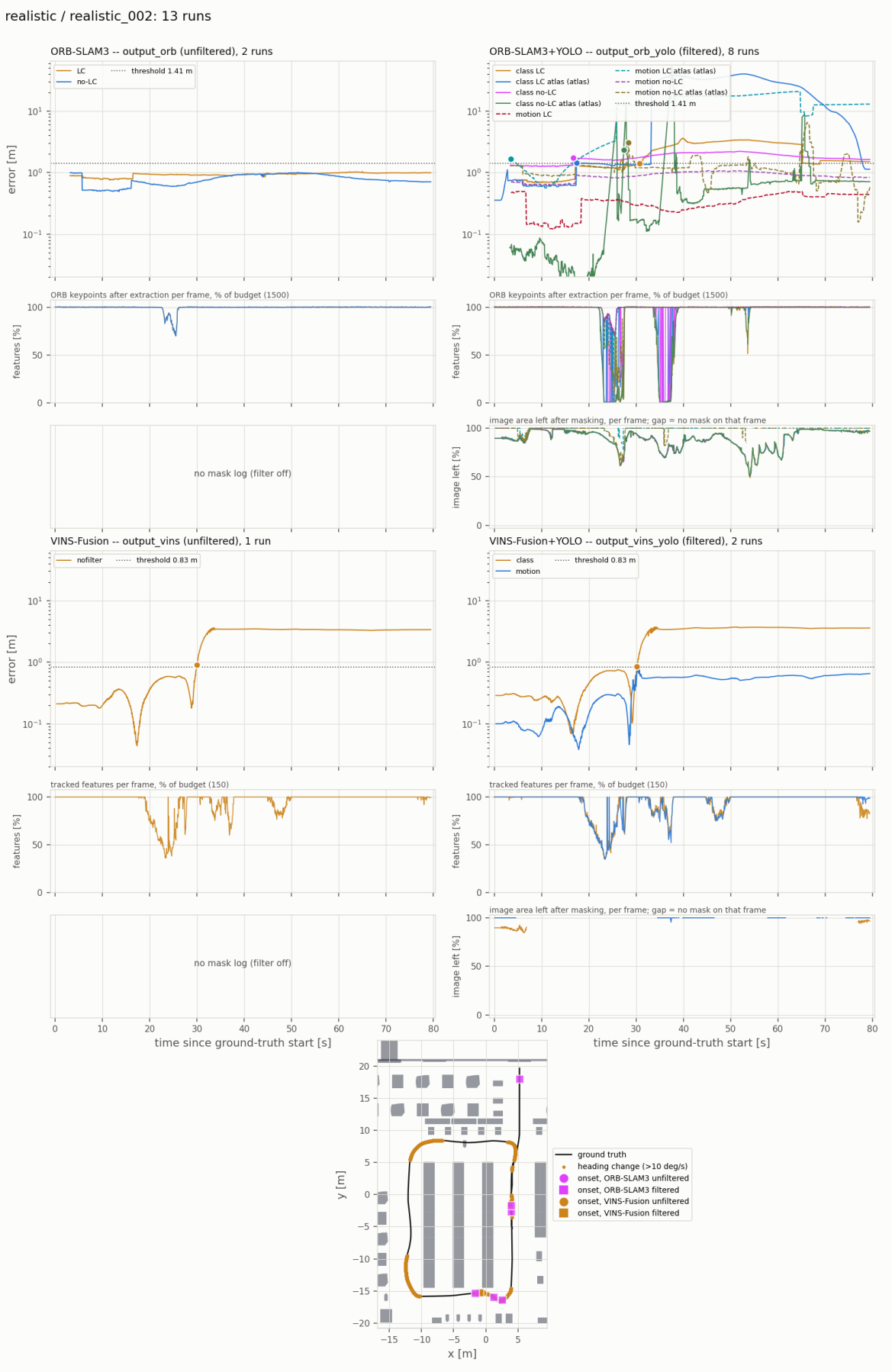


Figure 3.16: Realistic world, `realistic_002`, every run; panels as in figure 3.5. VINS-Fusion unfiltered crosses at 30 s in the heading change at 39 % of the route.

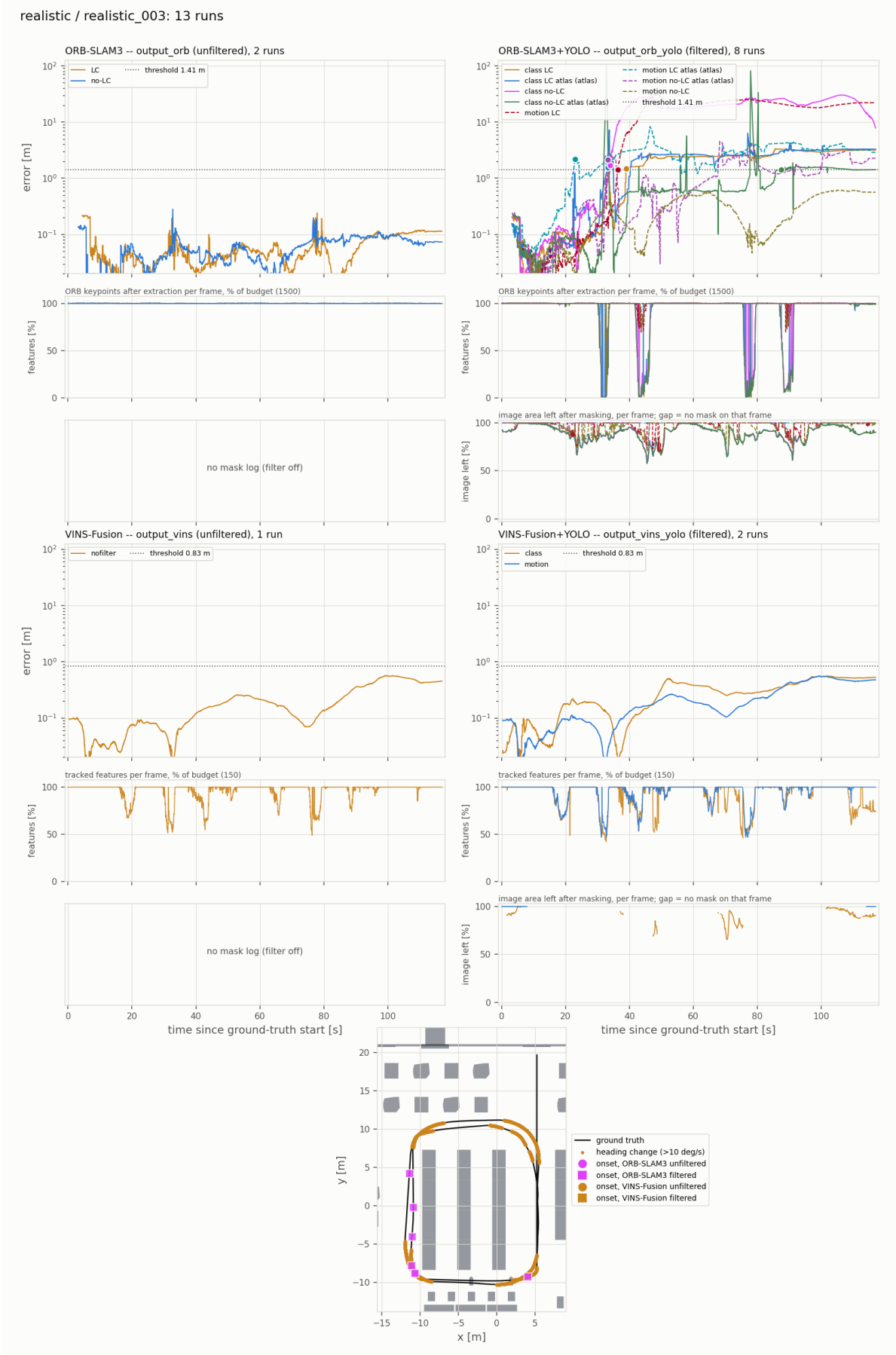


Figure 3.17: Realistic world, `realistic_003`, every run; panels as in figure 3.5. The class-masked ORB-SLAM3 runs cross at or after the first heading change following the long aisle, at 33–87 s, with the mask present on nearly every frame; the unfiltered runs never cross.

4.3.1.3 Summary and future plan

Summary. The two testbeds answer different questions and neither answers the one the experiment was designed for. The extreme world shows what the estimators do when the scene is not physical, and that is withheld. The realistic world shows that warehouse traffic degrades rather than breaks tracking, and by how much, from single runs. The filtering hypothesis has had no admissible test in either: zero valid pairs in the realistic world, two in the extreme world that cannot be read because the testbed is invalid. The motion gate is a negative result in both: it rarely paints, so it costs throughput without changing what it was meant to change. Neither pipeline runs at input rate in either testbed. Why the error rises when it does, and the lever-arm correction behind this document's scoring, are analysed in `insights/experiment_3_insight/`.

Future plan. Repeat the three-way test on the realistic recordings with three runs per condition, every arm on the same node build, the baseline arm replayed through the same launch file as the filtered arms, and `detection_match_rate` and the processed-frame count checked against the gates before a run is kept. Raise the detector's delivered mask rate at rate 1.0 for VINS-Fusion, or state the deployment result as "the filter cannot run at this rate on this machine". Record a second route in each layout so that traffic and route are separable. Report the prior-atlas runs under relocalisation, where they belong.

4.3.2 Real robot

Status: partial. Four real recordings with an independent planar reference. The first, `dataset_realv6_000` (31 October 2026), carries five unfiltered runs whose raw data exist: ORB-SLAM3 stereo-inertial with GPU or CPU front end and loop closing on or off, and VINS-Fusion once. ORB-SLAM3 tracks the whole route in all four configurations at a provisional 0.46 to 1.03 m planar ATE, the GPU front end at 0.46–0.52 m; VINS-Fusion resets 217 times on IMU-stream gaps and has no valid run. No filtered run exists, the `realv6` tracker export fails the pre-registered validity rules, the three `realv7` exports pass them and all twelve ORB-SLAM3 runs on them score after a frame repair in the scorer (0.25–0.27 m on `000`), but the IMU stayed uninitialised for most of nine runs (an upstream map reset at the initialisation, then a starved re-initialisation), VINS-Fusion has no valid run, and the marker-to-IMU lever arm is unmeasured, so the filter hypothesis remains untested on the real robot and no headline is claimed.

Objective. Establish whether the two unfiltered pipelines track the real robot against an independent reference, as the precondition for a filtered-versus-unfiltered test on real imagery.

Process. The bag was replayed into each node at real time on the afternoon of 3 November 2026; no detector ran. ORB-SLAM3 used `config/wil/stereo_imu.yaml` with the repaired library (fork patches 11–14b, node and library built alike), the 3 November stereo extrinsic and `IMU.NoiseGyro` 0.1 with `IMU.NoiseAcc` 1.0, ten times the values that had been copied from the VINS file, in four configurations: GPU or CPU feature extractor, loop closing on or off. VINS-Fusion used the same file with the gravity norm of an August recording. Earlier runs on this bag were overwritten by these and are not reported. The optical tracker export was scored with the real-robot procedure of Experiment 1.

Dataset. 220 s, 6 601 stereo pairs at 30 Hz, RealSense IMU at 200 Hz on the inherited topic name, wheel-speed feedback, a figure-of-eight of about 27 m at a median 0.15 m s^{-1} on the Gensurv floor, no scripted moving objects. Tracker export: 277 samples, median spacing 0.6 s, 66 gaps above 1 s, one sample 5.2 m off the path at 184.9 s.

Reporting rules. `script/eval_real_gt.py`: rigid SE(2) alignment with scale fixed, lever arm set to zero and unmeasured, every value provisional. The export fails the tool's gap and step rules, so the tool labels each run *failed (ground truth invalid)*; the values below are diagnostics. All are scored with the one off-path sample removed, an edit recorded here and applied nowhere else; table 3.15 shows what that sample does to one run. The bag was not on the machine, so the time offset is pinned to the value the wheel-speed cross-correlation found on 2 November and the manifests record the substitution. Sources: `output/compare/real/realv6_1103pm_*/`, `output/compare/real/realv6-insights/`, run directories under `output/output_orb/real/dataset_realv6_000/` and `output/output_vins/real/dataset_realv6_000/`.

Results. ORB-SLAM3 follows the whole 218 s route in every configuration: 0.52 and 0.46 m with the GPU extractor (loop closing on and off), 0.62 and 1.03 m with the CPU extractor, which drops two thirds of the frames at real time (table 3.14, figure 3.18). VINS-Fusion tracked 18 s before 217 IMU-gap resets took over (table 3.17) and has no entry in the accuracy table.

Table 3.14: ORB-SLAM3 on `dataset_realv6_000` against the cleaned tracker truth, 3 November 2026 afternoon series: repaired library, replay rate 1.0, 3 November stereo extrinsic, `IMU.NoiseGyro` 0.1 and `IMU.NoiseAcc` 1.0, GPU or CPU feature extractor, loop closing (LC) on or off. Aligned planar ATE after a rigid SE(2) fit with scale fixed; RPE at 1 s translational; path ratio = estimate / truth; resets = `local_mapping_bad_imu` map resets; dropped = camera frames the node's queue discarded of about 6 480 received. Lever arm tracker-marker to IMU unmeasured (zero applied); time offset pinned to −4.14 s; all values provisional under the scorer's validity rules.

Run	ATE [m]	median [m]	max [m]	$RPE_{1\,s}$ [m]	path ratio	resets	dropped
ORB-SLAM3, GPU, LC on	0.52	0.44	1.11	0.057	0.85	2	2420
ORB-SLAM3, CPU, LC on	0.62	0.48	1.38	0.110	0.88	2	4363
ORB-SLAM3, GPU, LC off	0.46	0.36	0.99	0.056	0.87	0	3201
ORB-SLAM3, CPU, LC off	1.03	0.89	1.83	0.223	0.96	3	4204

Table 3.15: The same ORB-SLAM3 run (GPU, LC on) scored against the tracker export as delivered and with one sample removed (the 5.2 m excursion at 184.9 s). The time offsets are the values the wheel-speed cross-correlation found on each export on 2 November (peak correlation 0.10 raw, 0.61 cleaned), applied here as pinned values. The cleaned truth still fails the gap rule (66 gaps above 1 s, tracker at 1.7 Hz).

Truth	GT path [m]	offset [s]	ATE [m]	median [m]	path ratio
as delivered	36.8	4.38	1.174	0.851	0.65
one sample removed	27.1	-4.14	0.519	0.437	0.85

Analysis. The front end is what separates the runs. The GPU extractor drops a third to a half of the frames and holds 0.5 m; the CPU extractor drops two thirds and loses a factor of two, one of its runs with a 2 m vertical excursion the tool flags as implausible and the other with three resets and a half-second of lost tracking. Loop closing changes nothing that the front end does not change more. The remaining resets are an initialisation effect, two or three `local_mapping_bad_imu` map resets inside the first 13 s and none after (zero in the GPU run without loop closing); after them the error is a slow wave between 0.05 and 1 m, not an accumulating drift (figure 3.19), and it does not rise while turning in any run (table 3.16). The estimates are a uniform 12–15 % smaller than the tracker's path, which the unmeasured lever arm, a stereo-baseline error or a tracker scale error could each explain. One tracker sample decides the headline: scored against the export as delivered, the GPU run reads 1.17 m with a path ratio of 0.65, because that sample adds 10 m of false path and dominates the time alignment (table 3.15).

The setting was reached by elimination. With the IMU noise densities copied from the VINS file the estimator was unstable on this recording, running away in its velocity state at a different place in each run;

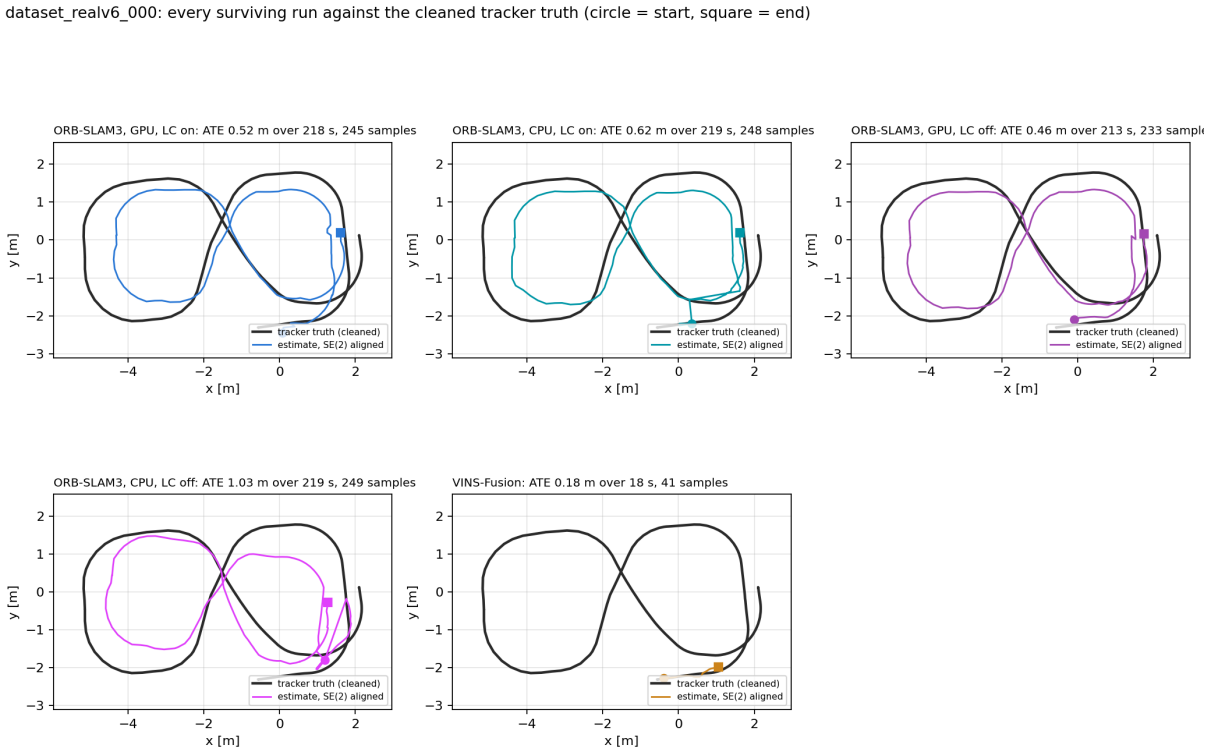


Figure 3.18: Every run against the cleaned tracker truth after rigid SE(2) alignment, one panel per run: ORB-SLAM3 with GPU or CPU extractor and loop closing on or off, and VINS-Fusion's 18 s.

Table 3.16: ORB-SLAM3 on `dataset_realv6_000`, cleaned truth: growth rate of the planar error while the aligned estimate's heading changes faster than $10\,^\circ\,\mathrm{s}^{-1}$ and on straights, with the time in each state. Positive = error growing. One run per configuration, so no test is possible.

Run	turning [mm/s]	straight [mm/s]	time turning [s]	time straight [s]
ORB-SLAM3, GPU, LC on	-11.2	-0.3	23	171
ORB-SLAM3, CPU, LC on	-29.8	-1.4	23	162
ORB-SLAM3, GPU, LC off	4.9	-2.1	11	185
ORB-SLAM3, CPU, LC off	-49.9	-0.9	20	182

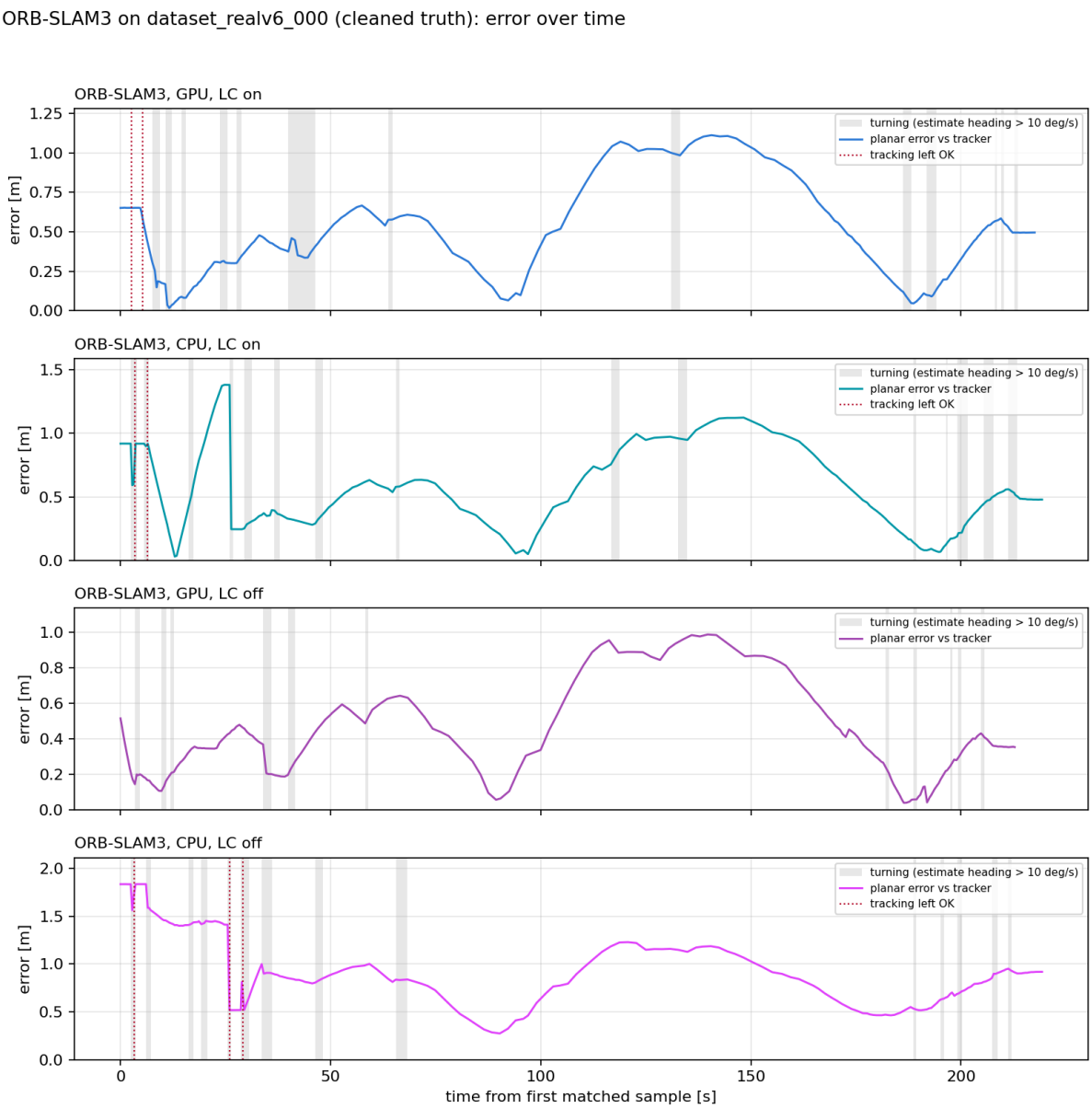


Figure 3.19: ORB-SLAM3 planar error over the run, one panel per configuration; grey bands are turns, dotted red the initialisation resets.

six out-of-bounds reads in the library were fixed without removing it, every other candidate was varied without removing it, and raising the IMU noise densities tenfold removed it in every run since. Those earlier runs were overwritten and carry no number here; the search is in the ORB-SLAM3 engineering record.

VINS-Fusion's failure is a delivery defect. The node received 3 187 of 6 601 frames but used 6 498 of the 43 494 IMU messages it received, reset 217 times on `imu_gap`, and stopped after 19 s; behind that defect the configuration still fixes a gravity norm of 9.2642 against the 9.878 m s^{-2} this IMU reads at rest.

Table 3.17: VINS-Fusion on `dataset_realv6_000`, 3 November 2026 (`config/wil/stereo_imu.yaml`, IMU noise ×10, gravity norm still 9.2642 against the 9.878 m s^{-2} this IMU reads at rest). The scored span is the 18 s it tracked before its last reset and is not a result.

Quantity	Value
camera frames received (bag: 6 601)	3187
IMU messages received / samples used	43494 / 6498
`imu_gap` resets	217
poses written / span of poses	286 / 19 s
tracker samples matched	41 of 43 in span
ATE over the tracked span	0.18 m (not a result)

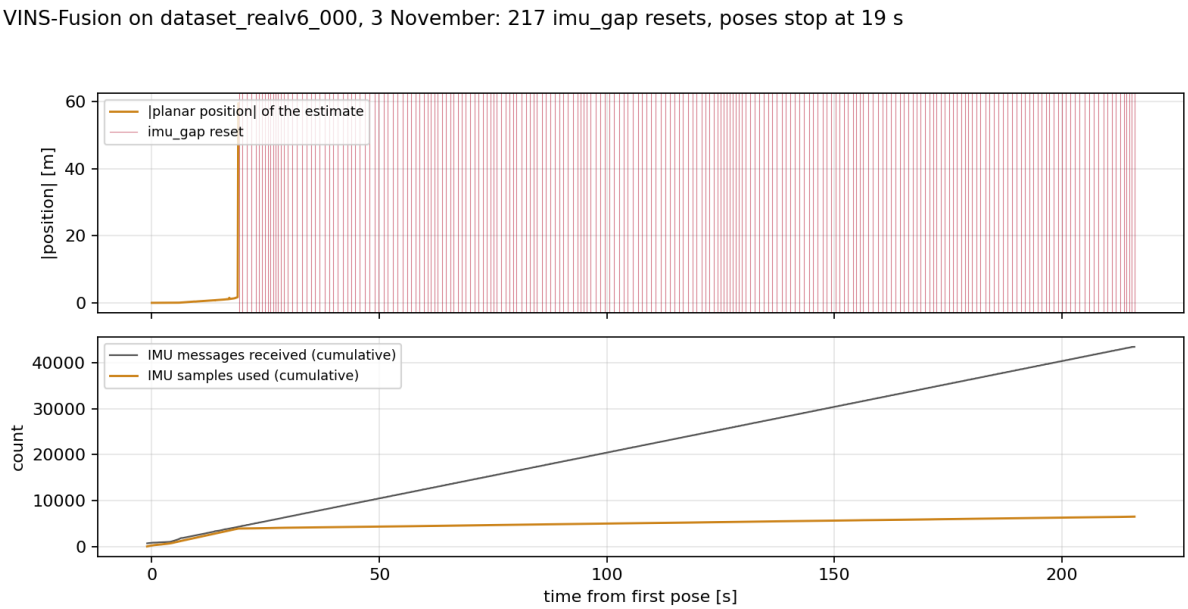


Figure 3.20: VINS-Fusion: distance from origin with every `imu_gap` reset marked, and IMU messages received against samples used.

Second recording set, realv7. Three further recordings of 3 November 2026 (`dataset_realv7_000`, `_001`, `_002`: 21.6, 31.7 and 23.4 m over 97, 134 and 104 s) have tracker exports that pass every validity rule (5–6 Hz, no gap, no step). Each carries the same four ORB-SLAM3 configurations and one VINS-Fusion run. As scored on 3 November (table 3.18, figures 3.21 and 3.22), a plane fitted to the ORB-SLAM3 estimate lies within 10° of horizontal in three runs of twelve and at 88–90° in five, every `000` run among them, with the plane normal exactly a body axis in every run.

The per-frame diagnostics give the mechanism. The IMU initialised 3–4 s after the first pose. The upstream rule that resets a map with less than 2 cm between consecutive keyframes in the first ten seconds after the initialisation then discarded it, the robot being still at rest. The new map starts in the camera

frame of the reset instant, and in it the re-initialisation was starved: upstream attempts it only with an empty keyframe queue, which a keyframe every 0.25 s and 250 ms of map-point creation per keyframe never allowed. The IMU re-initialised after 37–92 s in seven runs, never in two, and early (6.7 s) in the one run that was level. Until then the estimate is stereo-only in a map whose vertical is the camera's y axis, and the late re-initialisation rotates the map under the already-written poses (the 1.9–6.8 m steps).

The scorer repairs both without the truth: the pre-initialisation poses are rotated by the rotation measured at the initialisation frame, and each map's fitted plane is levelled. With the bags on the dataset drive the time offsets come from the wheel speed on all three recordings (table 3.19, figure 3.23). `000` scores 0.25–0.27 m in all four configurations over 89 s (RPE at 1 s 0.05 m), with the IMU initialised for at most a quarter of the frames. `001` scores 0.44–0.82 m over 126 s, one run failing the scorer's one-metre vertical-range rule by two centimetres. `002` scores 0.47 m for the CPU front end with loop closing; two runs lose tracking inside the route and one drifts (vertical range 4.3 m, failed). The init-time resets leave every other run *partial* under the scorer's rules.

The fork's patch 15 adds two switches that attempt the initialisation with a queued backlog and skip the low-motion reset for stereo; they are on in the real-rig configuration and not yet exercised on a recording. VINS-Fusion diverges on all three with the IMU stream intact (zero resets) and 40–45 % of the frames received (table 3.20); its gravity norm has since been set to the measured 9.886 m s^{-2}.

Table 3.18: ORB-SLAM3 on the three `dataset_realv7` recordings, 3 November 2026 afternoon, same setting as the realv6 runs. "Tilt" is the angle between the plane fitted to the estimate's 3-D positions and the horizontal; "off-plane" the rms distance of the positions from that plane. A floor robot's estimate should be horizontal: a tilt near 90° means the poses were written while the map was not yet gravity-aligned (IMU not initialised: the vertical is then the camera's y axis), see table 3.19 for the initialisation history and the repaired scores. ATE is planar after SE(2) alignment against the raw tracker export (which passes the validity rules on all three recordings); "offset" says how the time offset was found: *wheel* = wheel-speed cross-correlation with the bag present (realv7_000), *estimator* = the estimate's own speed with the bag absent (uncertain; superseded by the wheel offsets of table 3.19). Lever arm unmeasured (zero).

Recording	Run	tilt [°]	off-plane [m]	z range [m]	ATE [m]	median [m]	path ratio	resets	offset
`realv7_000`	GPU, LC on	89.1	0.35	6.87	2.26	2.08	0.55	None	wheel
`realv7_000`	GPU, LC off	89.5	0.02	6.98	2.32	2.19	0.47	None	wheel
`realv7_000`	CPU, LC on	88.8	0.41	6.88	2.24	2.04	0.56	None	wheel
`realv7_000`	CPU, LC off	87.9	0.58	6.79	2.19	1.87	0.67	1	wheel
`realv7_001`	GPU, LC on	88.8	0.05	9.64	3.10	3.17	0.66	2	estimator
`realv7_001`	GPU, LC off	6.1	1.12	5.77	1.44	0.95	1.19	2	estimator
`realv7_001`	CPU, LC on	1.5	0.07	0.38	0.31	0.23	1.09	1	estimator
`realv7_001`	CPU, LC off	13.0	1.05	5.23	1.48	0.79	1.16	1	estimator
`realv7_002`	GPU, LC on	37.7	0.87	4.08	1.94	1.46	1.15	3	estimator
`realv7_002`	GPU, LC off	59.7	0.83	4.80	2.11	1.76	1.04	2	estimator
`realv7_002`	CPU, LC on	5.2	0.73	2.94	0.85	0.64	1.06	1	estimator
`realv7_002`	CPU, LC off	21.7	0.84	7.03	1.67	1.19	1.61	0	estimator

[Sparse reference] The tracker ran at 1.7 Hz rather than its nominal 7 Hz on `realv6`, and the validity rules were written for the faster export. Either the export has to be re-captured at full rate or a rule for sparse references has to be written before any value from this recording is quoted outside a diagnostic.

Summary. The real robot has an unfiltered ORB-SLAM3 reference at a provisional 0.46–0.52 m on `realv6` (GPU front end) and 0.25–0.27 m on `realv7_000` with valid truth after the scorer's frame repair, a measured cost of the CPU front end at real time, the cause of the IMU staying uninitialised for most of nine `realv7` runs (an upstream map reset at the initialisation and a starved re-initialisation) patched in the fork but not yet exercised, and two diagnosed VINS-Fusion defects, one of them (the gravity norm) now corrected in the configuration. The experiment's own question, whether filtering helps, is still untested on real imagery.

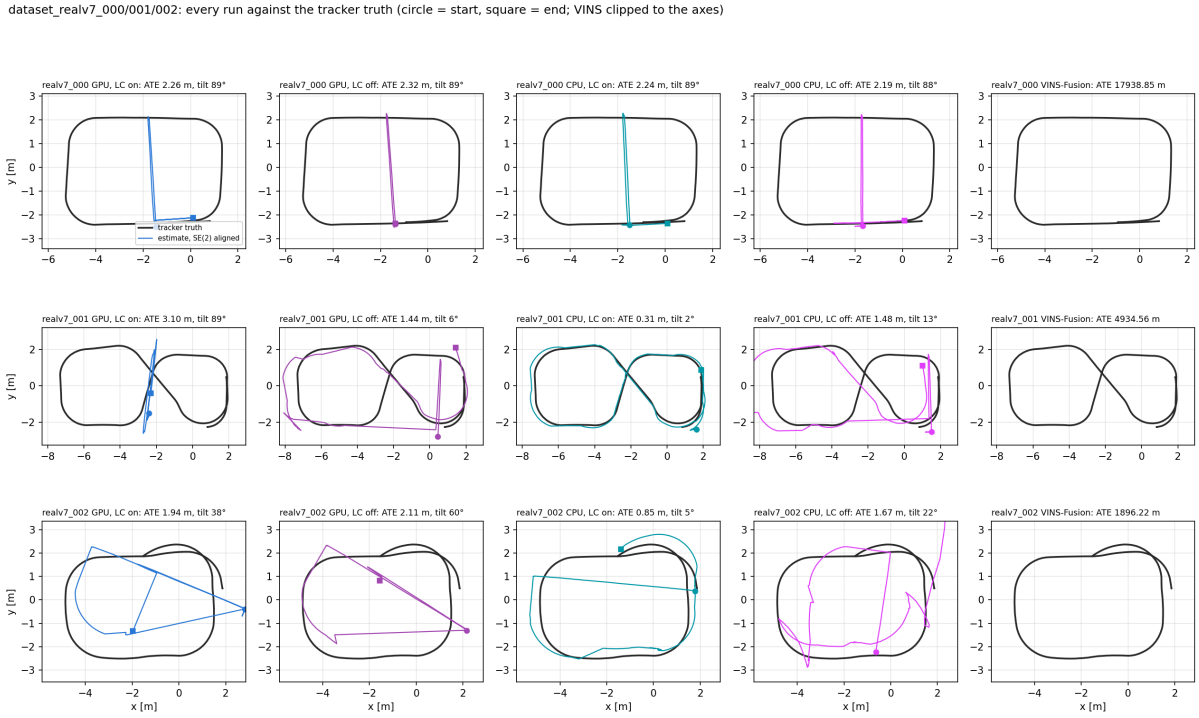


Figure 3.21: Every `realv7` run against its tracker truth after rigid SE(2) alignment: three recordings by four ORB-SLAM3 configurations and VINS-Fusion.

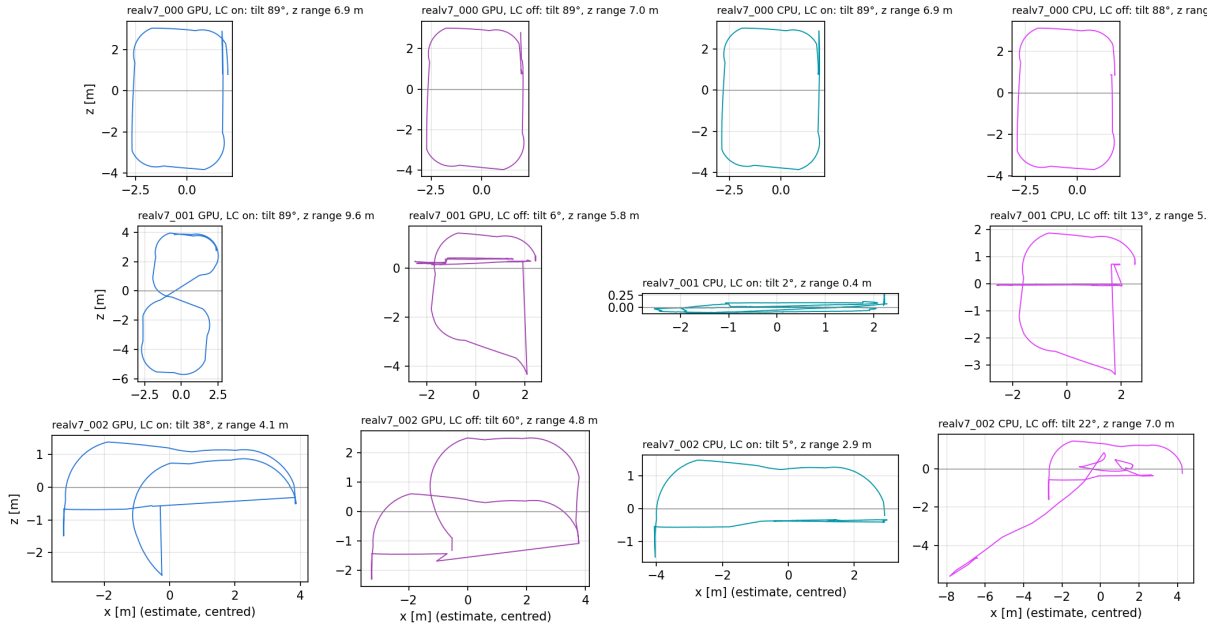


Figure 3.22: The twelve `realv7` ORB-SLAM3 estimates seen from the side (x against z, centred); a floor robot's path should be a flat line.

Table 3.19: ORB-SLAM3 on the three `dataset_realv7` recordings (3 November 2026, same setting as realv6), scored on 4 November with the bags on the dataset drive (wheel-speed time offsets) and the two frame repairs of `script/eval_real_gt.py`: the poses written before each IMU initialisation rotated into the frame used after it (`-join-imu-init`), and each map's fitted plane levelled (`-level-plane`). "IMU" is the share of tracked frames with the IMU initialised; "resets" the active-map resets after the first pose, in seconds (a reset restarts the position at the origin and is not stitched; those inside the route are tracking losses); "tilt" the plane tilt of the raw log, zero after the repair in every run. ATE is planar after SE(2) alignment, RPE over 1 s. Status from the scorer's rules ("partial" = a reset or loss the run recovered from). Lever arm unmeasured (zero); VINS-Fusion diverged on all three (table 3.20) and is omitted.

Recording	Run	IMU [%]	resets [s]	tilt [°]	ATE [m]	median [m]	$RPE_{1\,s}$ [m]	path ratio	status
`realv7_000`	GPU, LC on	6	3	89	0.25	0.24	0.05	1.05	partial
`realv7_000`	GPU, LC off	0	3	89	0.26	0.21	0.05	1.09	partial
`realv7_000`	CPU, LC on	11	5	89	0.27	0.21	0.05	1.06	partial
`realv7_000`	CPU, LC off	23	4	88	0.26	0.23	0.05	1.08	partial
`realv7_001`	GPU, LC on	3	4	89	0.48	0.47	0.07	1.07	partial
`realv7_001`	GPU, LC off	67	3, 5	6	0.82	0.63	0.15	1.15	partial
`realv7_001`	CPU, LC on	94	4	2	0.44	0.40	0.07	1.07	partial
`realv7_001`	CPU, LC off	65	5	13	0.56	0.50	0.10	1.11	failed
`realv7_002`	GPU, LC on	22	3, 49, 59	38	2.11	1.28	0.26	1.18	partial
`realv7_002`	GPU, LC off	22	49	60	2.08	1.83	0.23	1.29	partial
`realv7_002`	CPU, LC on	61	none	5	0.47	0.41	0.12	1.13	partial
`realv7_002`	CPU, LC off	62	none	22	1.67	0.98	0.70	1.68	failed

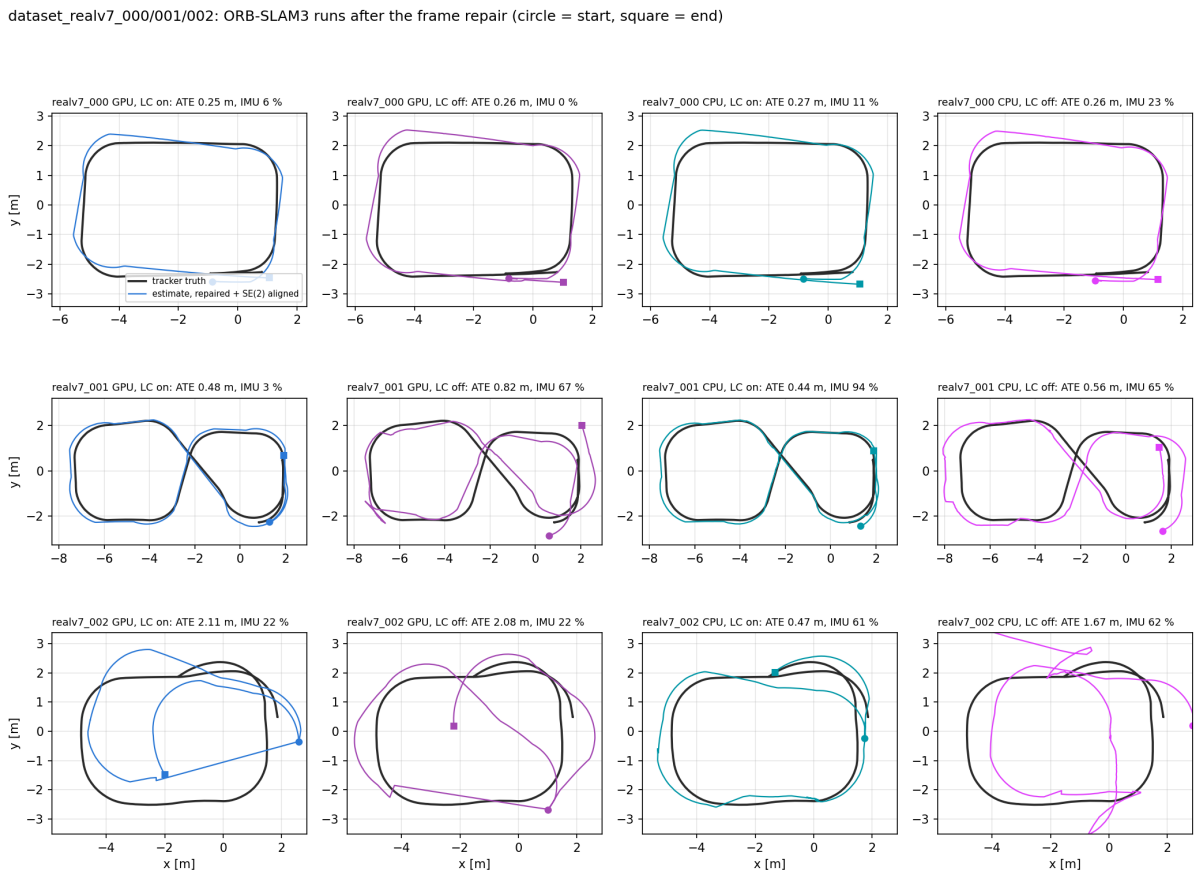


Figure 3.23: The twelve `realv7` ORB-SLAM3 runs after the scorer's frame repair, against the tracker truth after rigid SE(2) alignment; each panel's title gives the planar ATE and the share of tracked frames with the IMU initialised.

Table 3.20: VINS-Fusion on the three `dataset_realv7` recordings (`config/wil/stereo_imu.yaml`, IMU noise $\times 10$, gravity norm 9.2642). Frames received against the bag's count (30 Hz over the recording), IMU use, resets, and where the estimate ended.

Recording	frames received	IMU received / used	resets	z range [m]	$\lvert p \rvert$ at end [m]
`realv7_000`	1155 of about 2663	17906 / 19053	0	12834403	37554
`realv7_001`	1541 of about 3686	24809 / 26340	0	3701	15974
`realv7_002`	1271 of about 2910	19368 / 20627	0	25937	5252

Next action. Run one `realv7` recording with the fork's two initialisation switches on and confirm from the per-frame log that the IMU stays initialised from the first seconds; re-run VINS-Fusion with the corrected gravity norm; repeat each ORB-SLAM3 configuration under a new run name; measure the marker-to-IMU lever arm; then run both pipelines filtered and unfiltered on the same recording, with the detector live, and score them with the admissibility gates of the simulation slots.